

B.Tech. II Sem.
Physics-PH-113
“Diffraction of Light”

Fraunhofer Diffraction due to Single Slit & Double Slits

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Example of diffraction in real life:

1. Bending of light around the corners of door and windows.

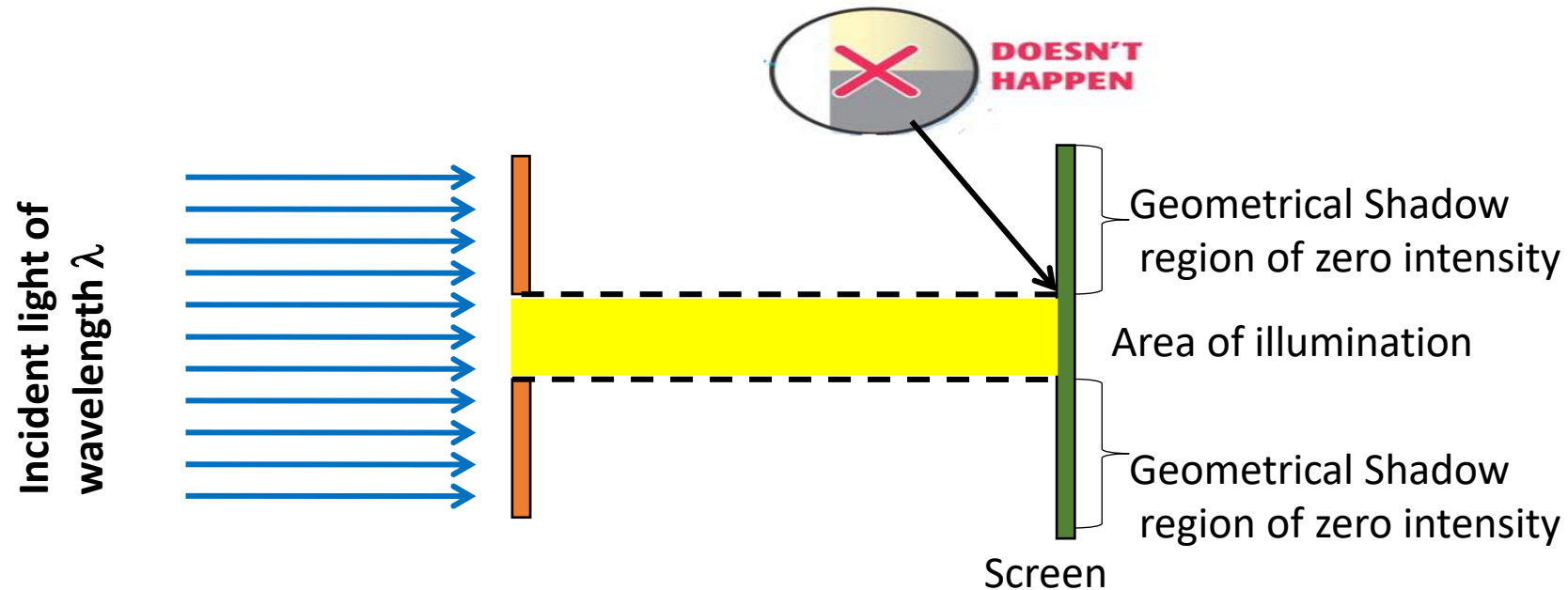


2. Sun rays coming from the clouds.



Diffraction of Light

According to geometric optics, a light source shining on an object in front of a screen should cast a sharp shadow. Surprisingly, this does not occur because of *diffraction*.

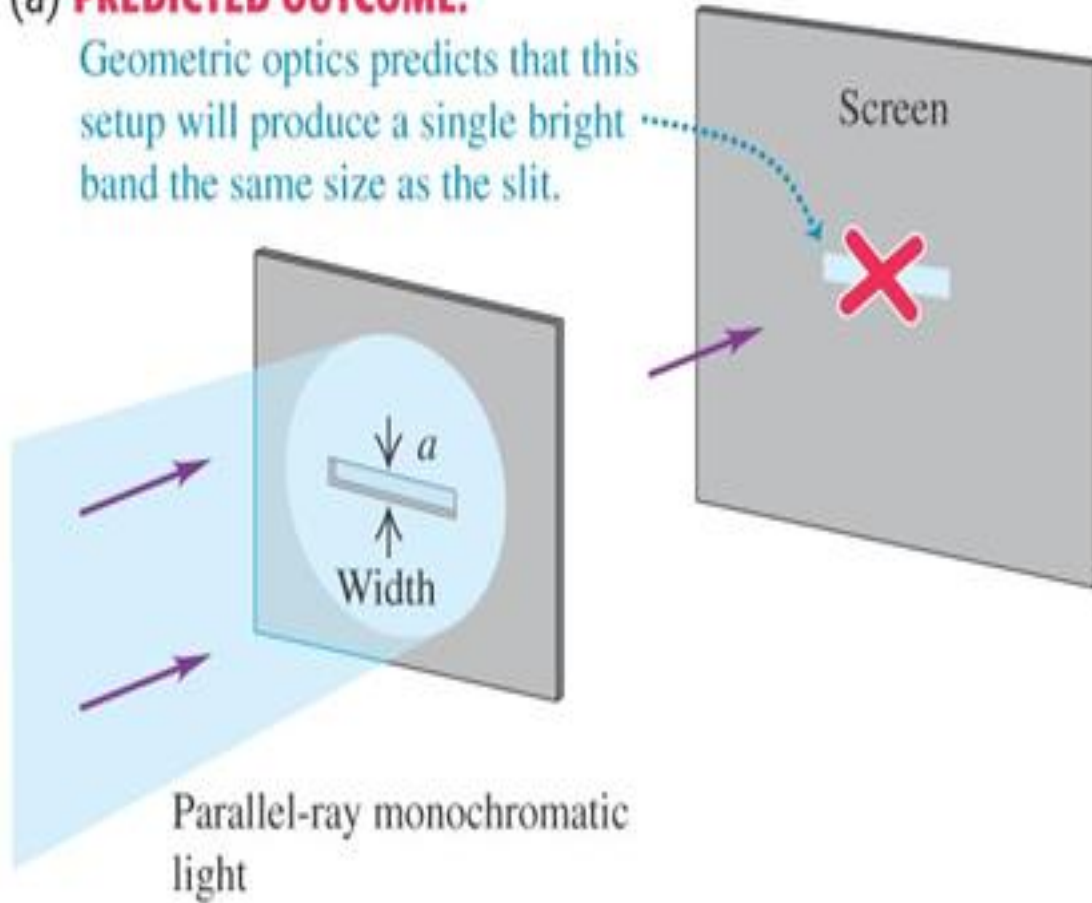


**Geometric optics predicts that this situation should produce a sharp boundary between illumination and geometrical shadow region.
That's NOT what really happens!**

Diffraction of Light

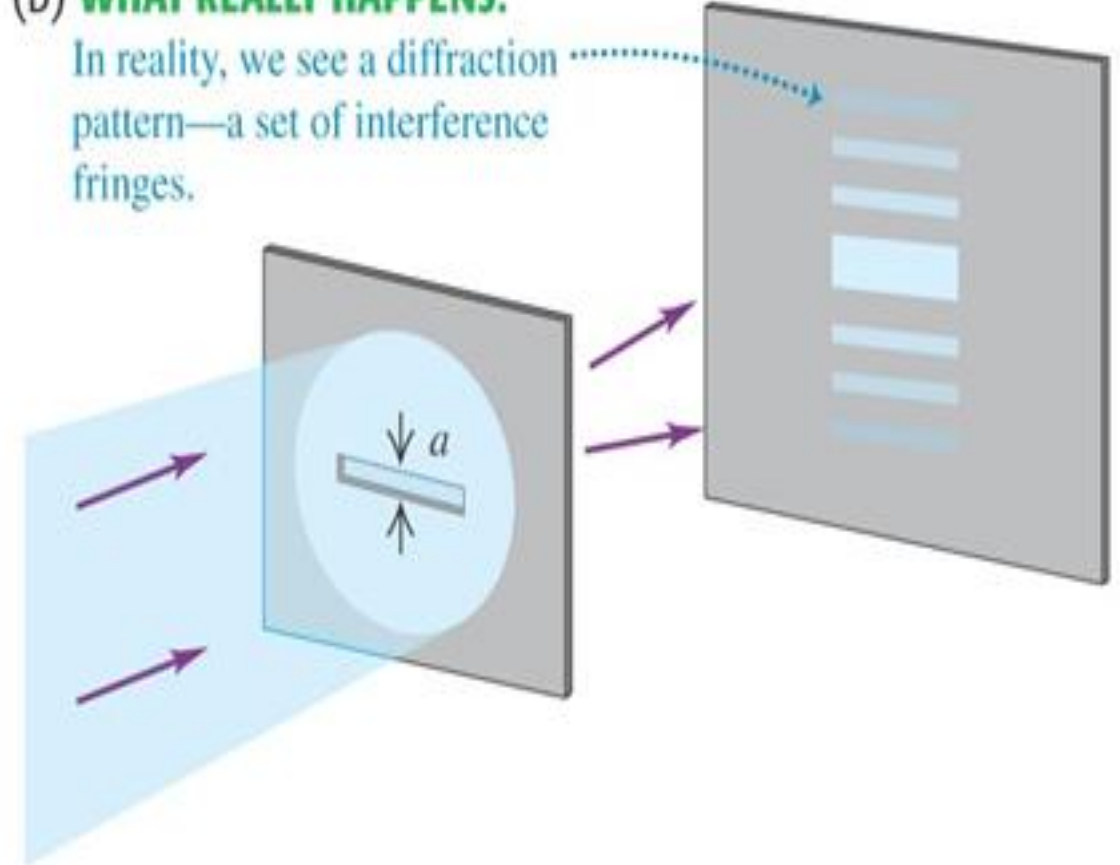
(a) **PREDICTED OUTCOME:**

Geometric optics predicts that this setup will produce a single bright band the same size as the slit.



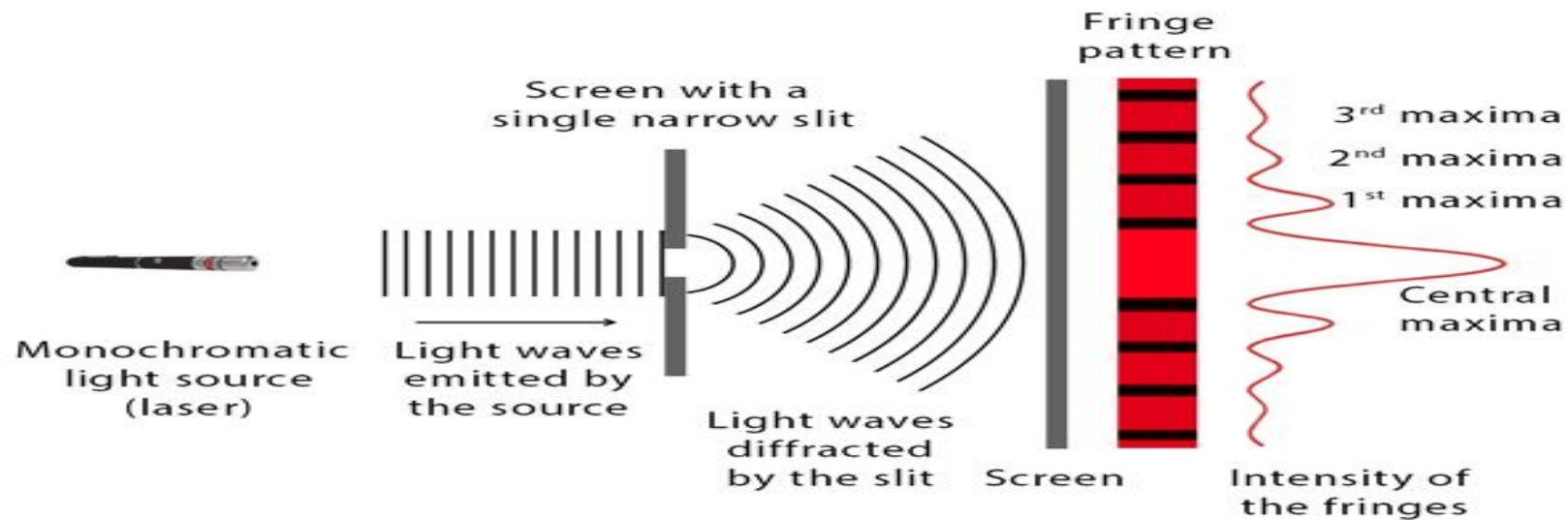
(b) **WHAT REALLY HAPPENS:**

In reality, we see a diffraction pattern—a set of interference fringes.



Diffraction of Light

- The phenomenon of bending of light around an obstacle or slit and spreading out in the geometrical shadow region is called diffraction.



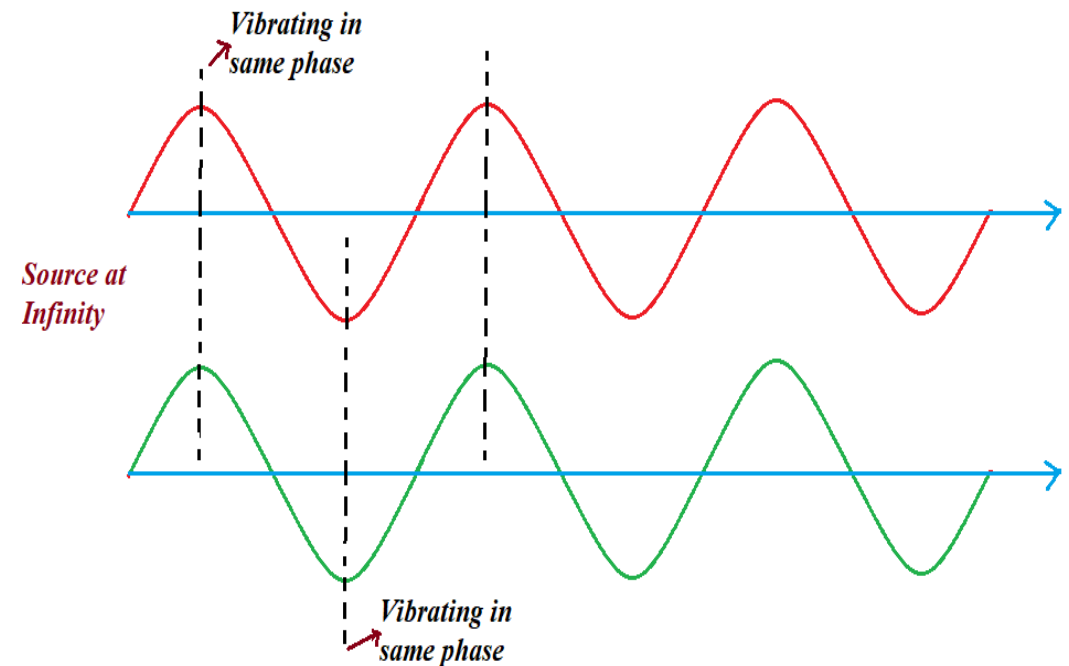
- It is due to superposition of secondary wavelets emitted from different points (infinite in number) of the same wavefront.

Wavefront

➤ **Wavefront:** A wavefront is defined as the continuous locus of all the particles of a medium, which are vibrating in the same phase..

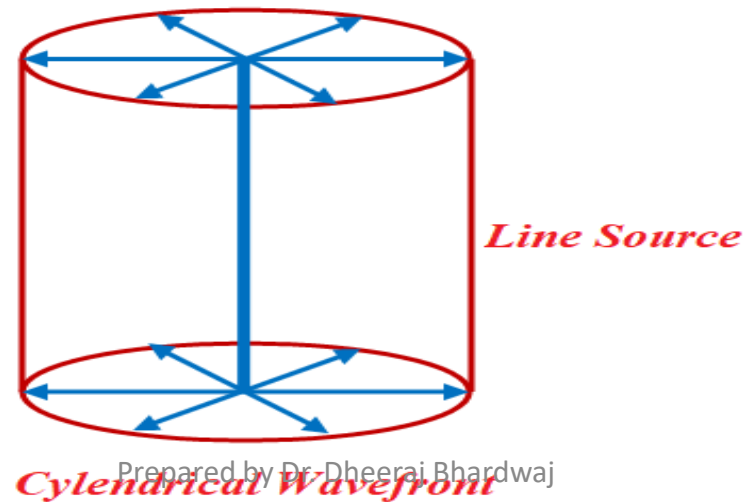
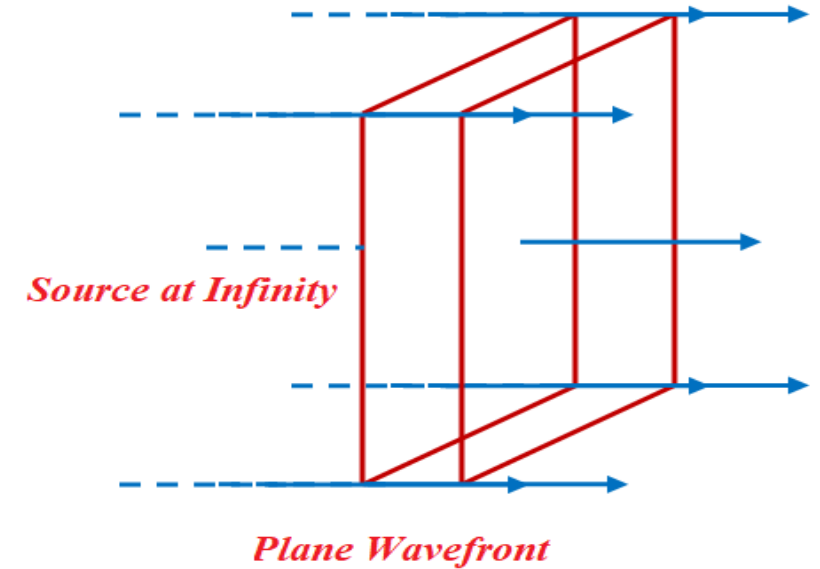
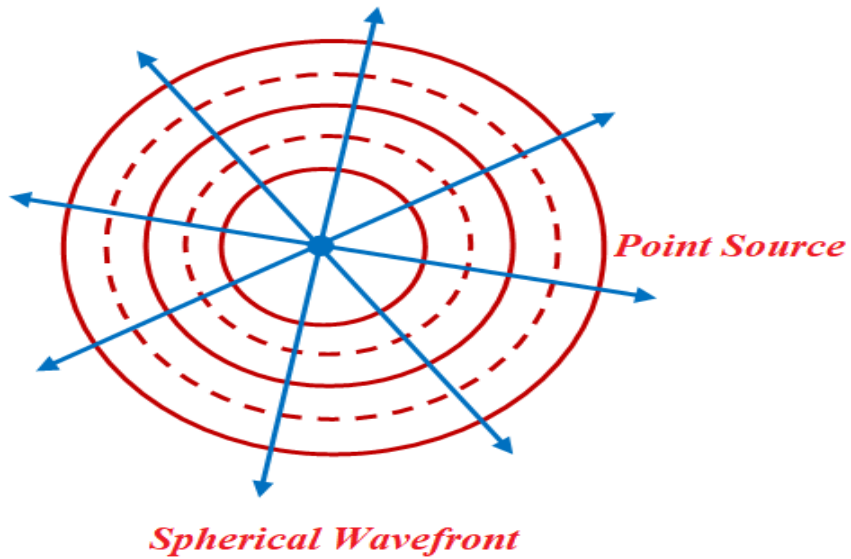


Same phase
Wavefront



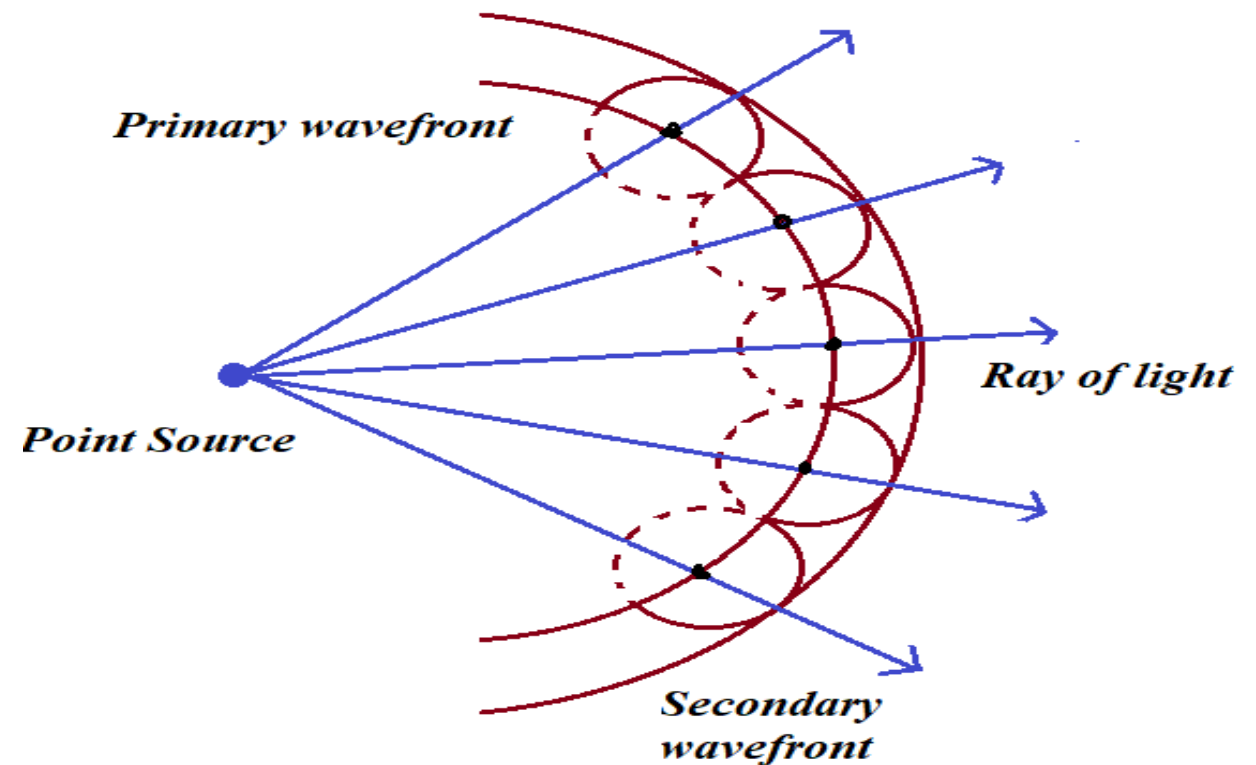
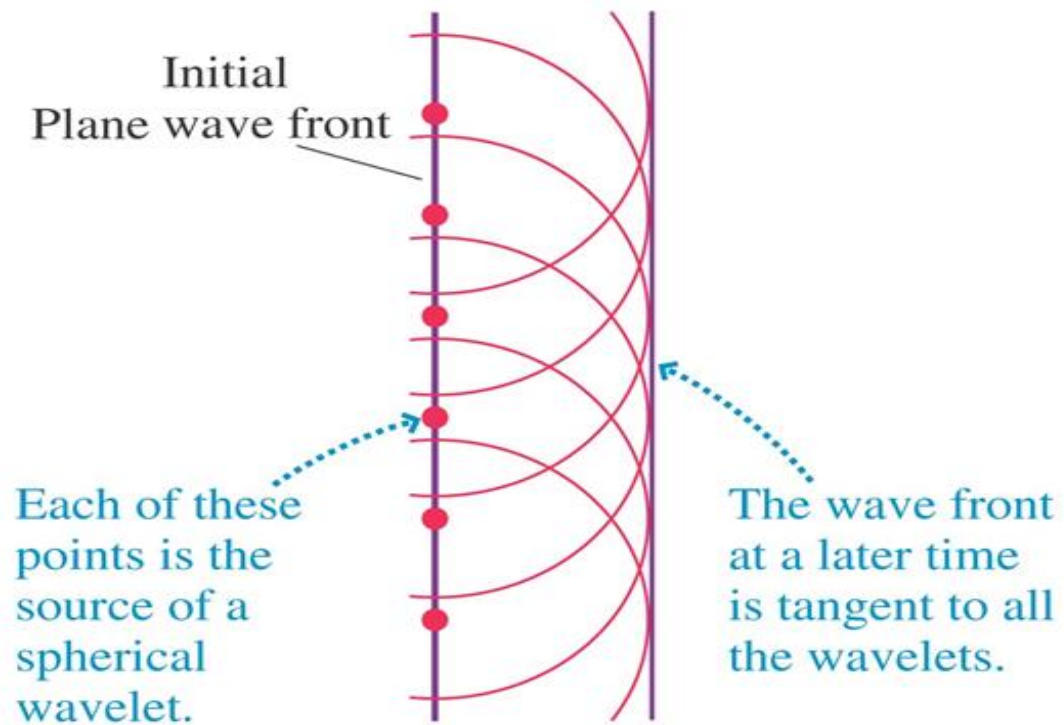
Wavefront

Types of wave fronts:



Huygens' Principle

- **Secondary Source** : According to Huygens' principle each point of the wavefront acts as a secondary source of light and sends out secondary wavelets in forward directions.



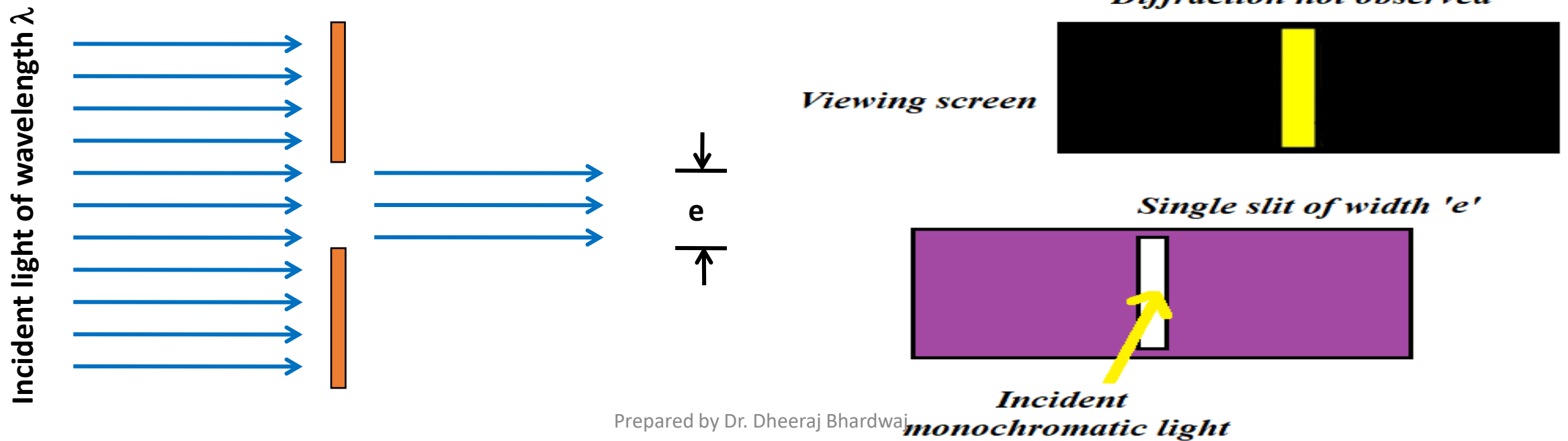
Diffraction of Light.....

Diffraction of light depends on the following -

- Size of the diffracting element (e)
- Wavelength of the light waves (λ)

Case – I

- When the size of diffracting element is large compared to a wavelength of light used ($e \gg \lambda$), the deviation or bending of wave is very small i.e. diffraction of wave is not observable.



Diffraction of Light.....

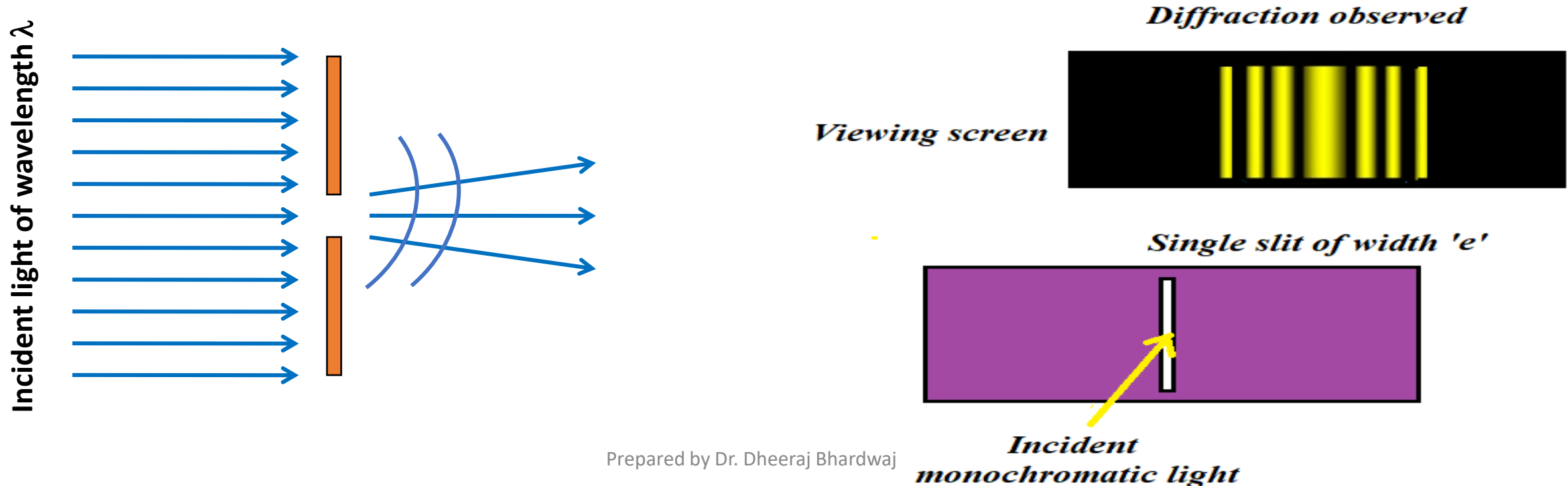


- If the size of the opening is large as compared to the wavelength of light, diffraction or bending of light is not noticeable.

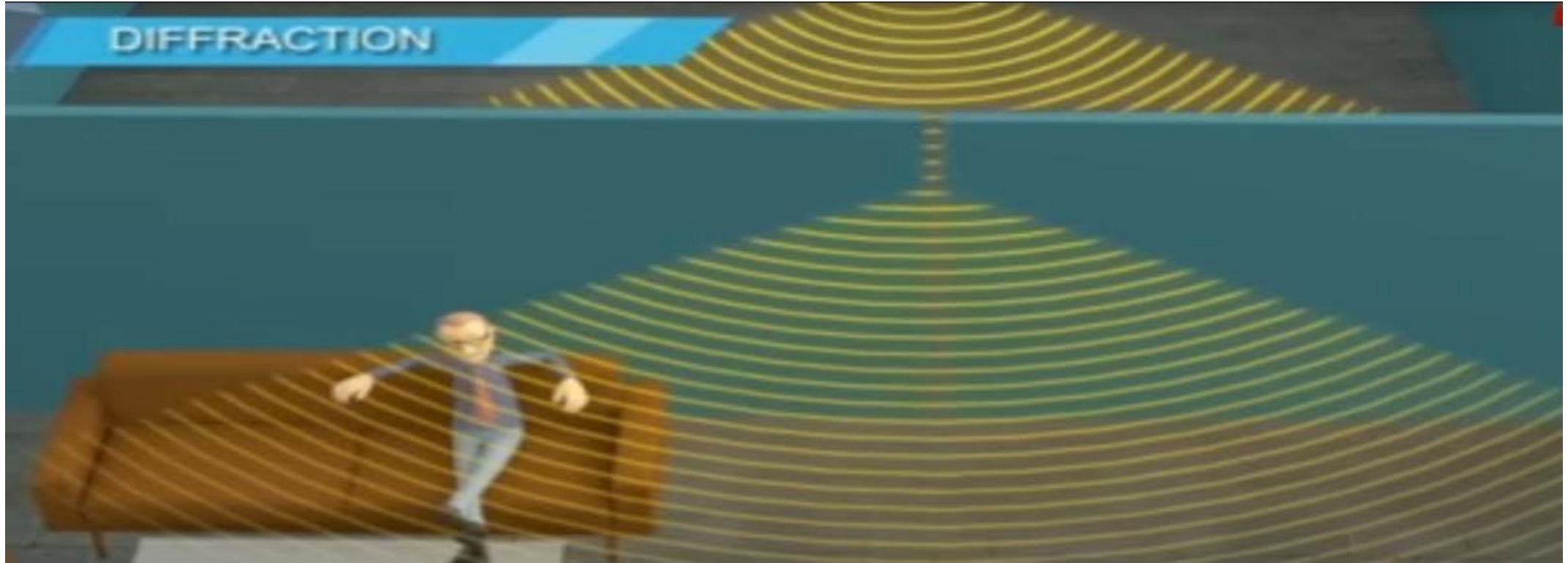
Diffraction of Light.....

Case—II

When the size of diffracting element is of the order of wavelength of light ($e \approx \lambda$), the deviation or bending of wave becomes much pronounced and this effect is noticeable.



Diffraction of Light.....

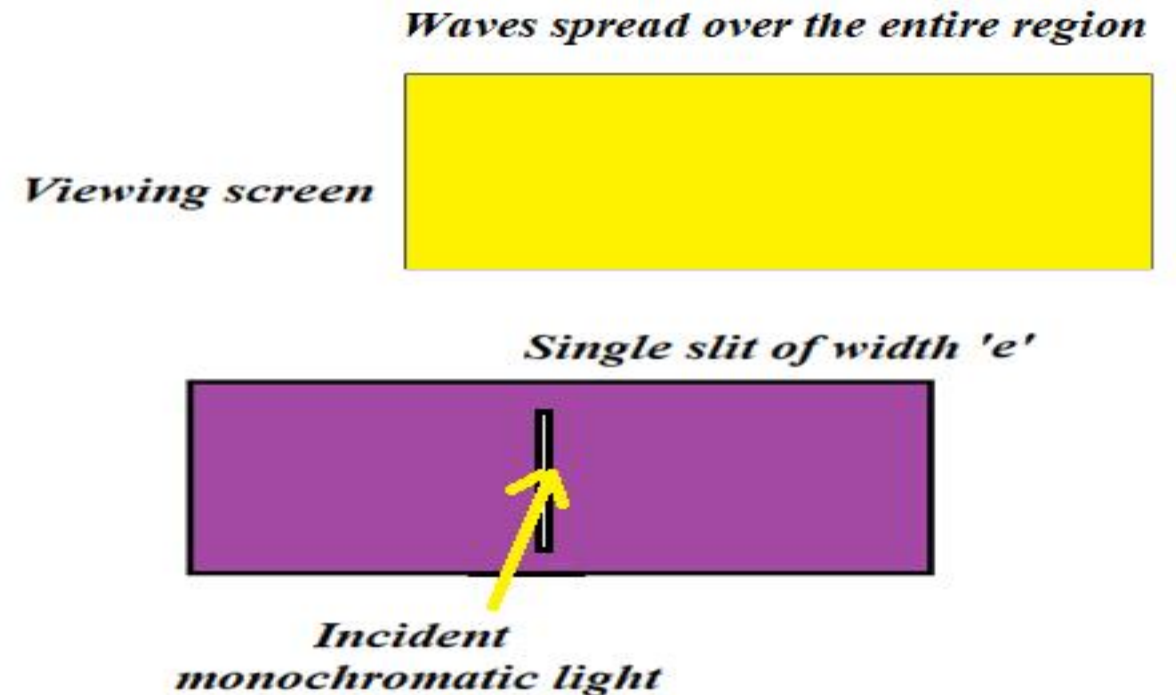
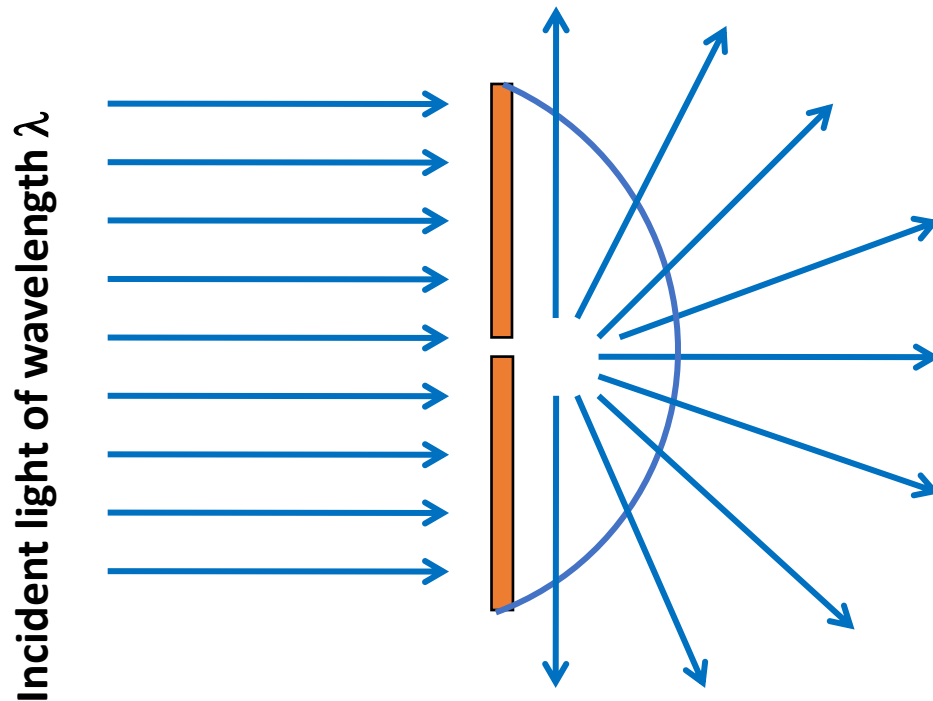


If the size of the opening is smaller or equal to the wavelength of light, diffraction is noticeable.

Diffraction of Light.....

Case-III.

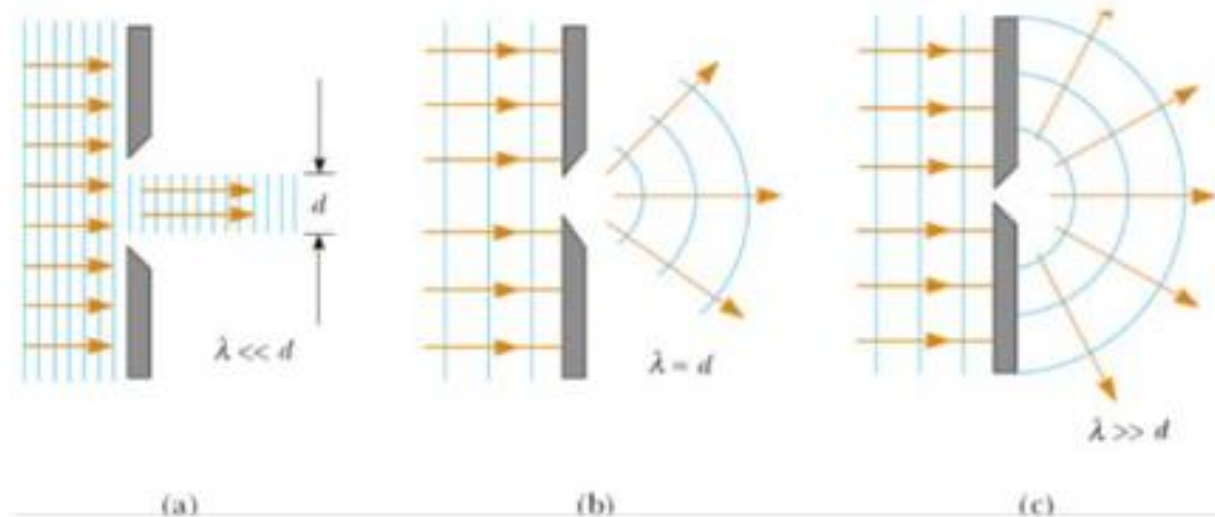
When the size of diffracting element is very small ($e \ll \lambda$), the waves spread over the entire region behind the obstacle or slit.



Diffraction

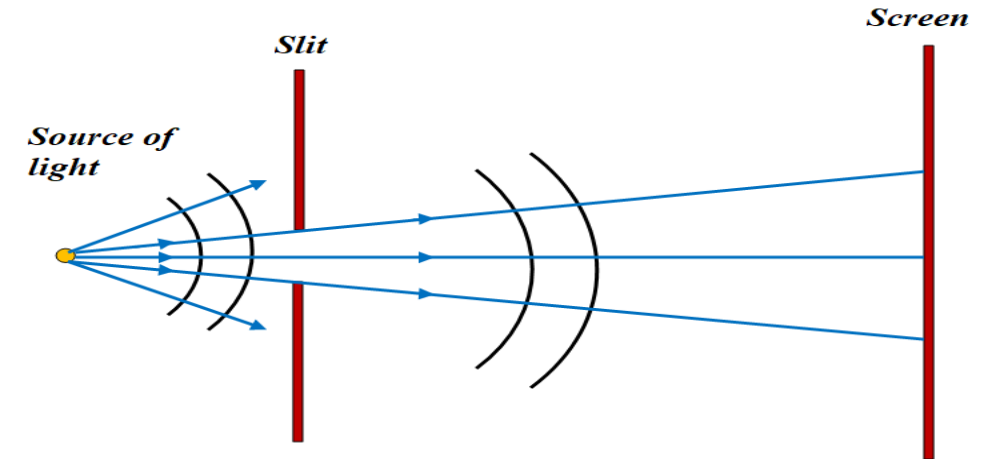
- Introduction to Diffraction Patterns

A plane wave of wavelength λ is incident on a barrier in which there is an opening of diameter d . (a) When $\lambda \ll d$, the rays continue in a straight-line path, and the ray approximation remains valid. (b) When $\lambda \approx d$, the rays spread out after passing through the opening. (c) When $\lambda \gg d$, the opening behaves as a point source emitting spherical waves.

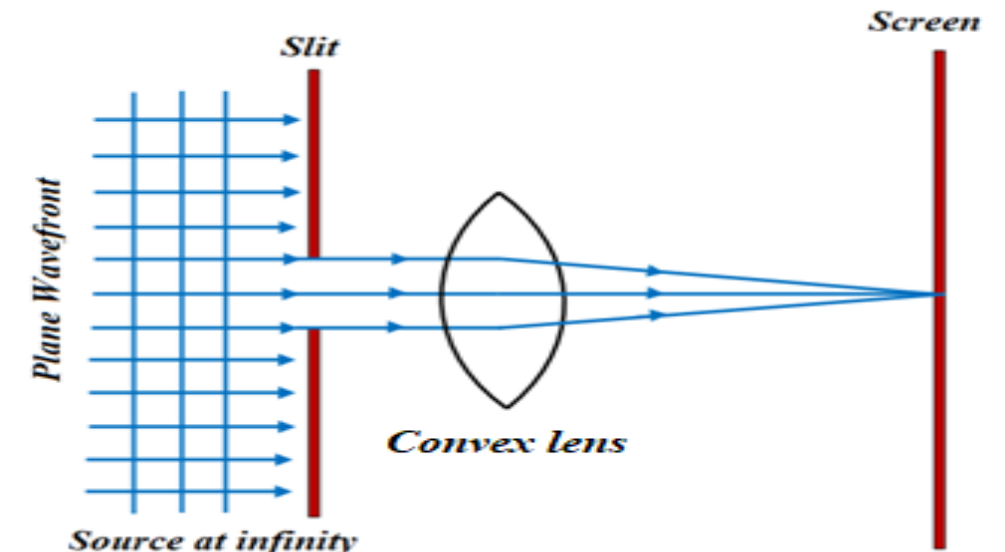


Types of Diffraction of light

I. Fresnel Diffraction: When the light source and the observation point are at finite distance from the obstacle or aperture, diffraction so produced, is called Fresnel diffraction. In this diffraction the incident and the diffracted wavefronts are spherical or cylindrical.

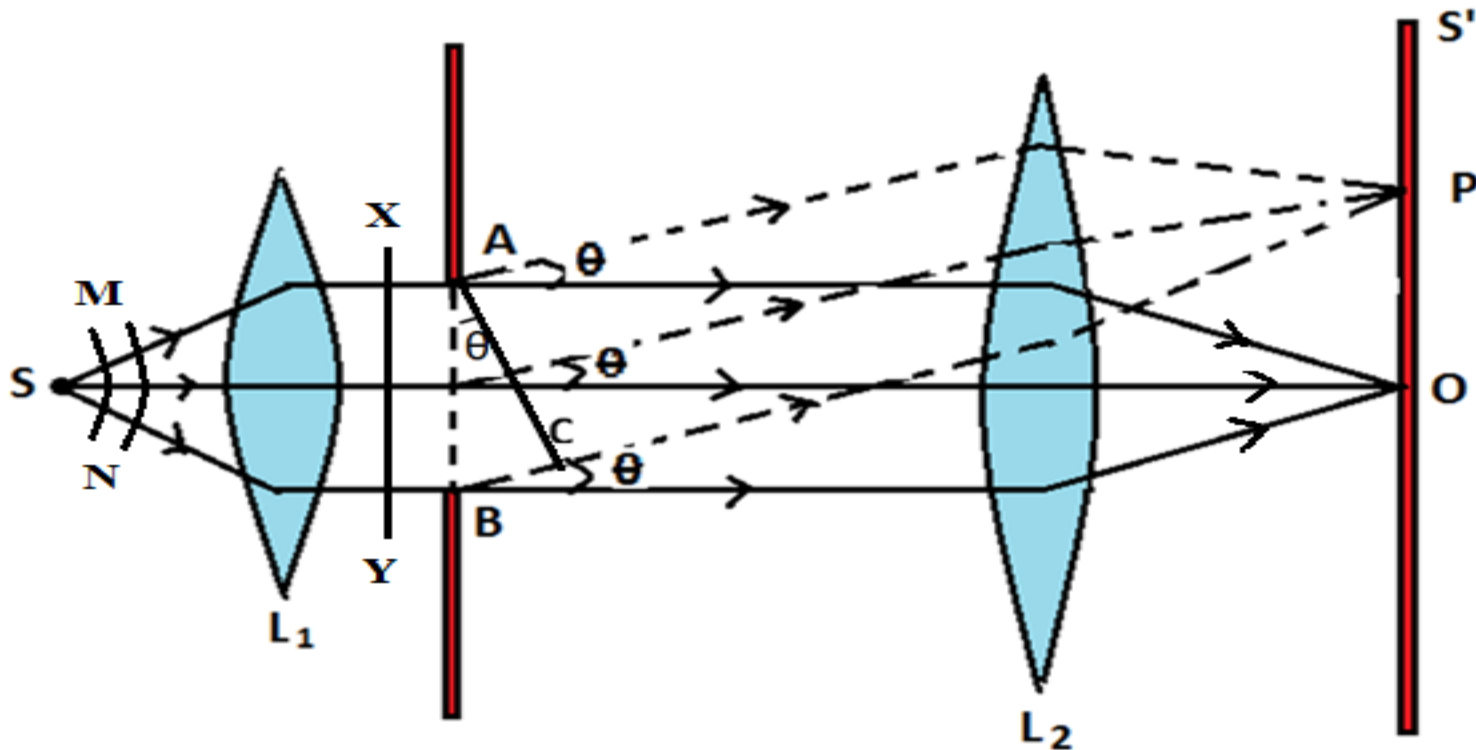


II. Fraunhofer Diffraction: When the light source and the observation point are at infinite distances from the diffractor, that is, the incident and the diffracted wavefronts are plane, diffraction so produced is called Fraunhofer diffraction.



Fraunhofer Diffraction	Fresnel Diffraction
The source and the screen are effectively at infinite distance from the diffracting element.	The source and the screen are at finite distance from the diffracting element.
Two convex lenses are used to observe diffraction effect.	Lenses are not used to observe diffraction effect.
Incident wavefront is plane.	The wavefront is either spherical or cylindrical.
The centre of diffraction pattern is always bright.	The centre of diffraction pattern may be bright or dark.
In this class of diffraction the angular inclinations are important.	In this class of diffraction the distances are important.
In this class of diffraction the effects of several diffracting elements can be added because theoretical analysis is much simple.	In this class of diffraction the effects of several diffracting elements cannot be added because theoretical analysis is complex.

Fraunhofer's Diffraction at a Single Slit



S :- Monochromatic light source

L_1 and L_2 :- Convex lenses

AB :- Slit of width of width ' e '

θ :- Angle of diffraction

S' :- Screen

O :- Position of central maximum

P :- Point of observation

Fraunhofer's Diffraction at a Single Slit....

The path difference between wavelets from A to B in a direction 'θ'

$$BC = AB \sin \theta \Rightarrow e \sin \theta \quad \dots(1)$$

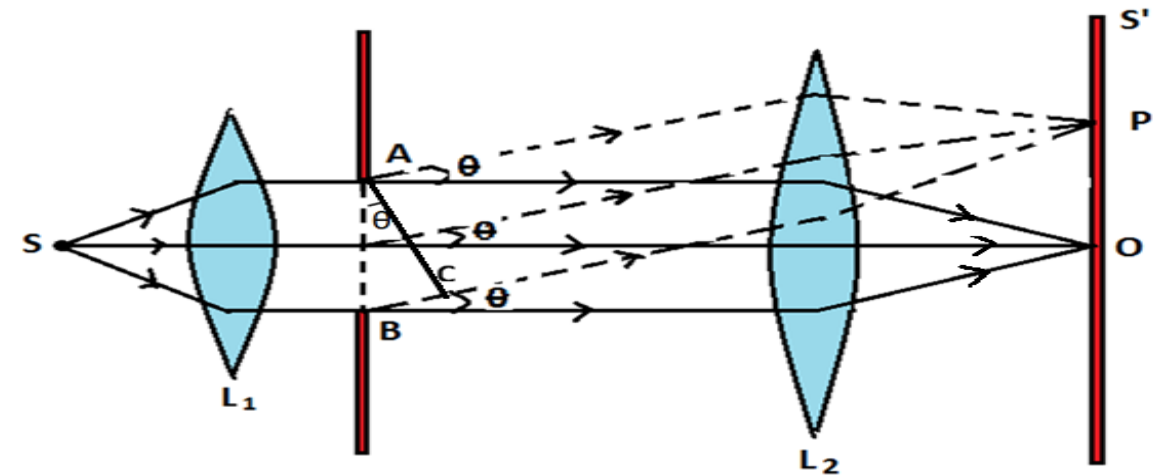
As Single slit width $AB = e$

From the relation between phase difference and path difference

$$\delta = \frac{2\pi}{\lambda} \times \text{path difference}$$

$$\delta = \frac{2\pi}{\lambda} \times e \sin \theta = 2\alpha(\text{let}) \quad \dots(2)$$

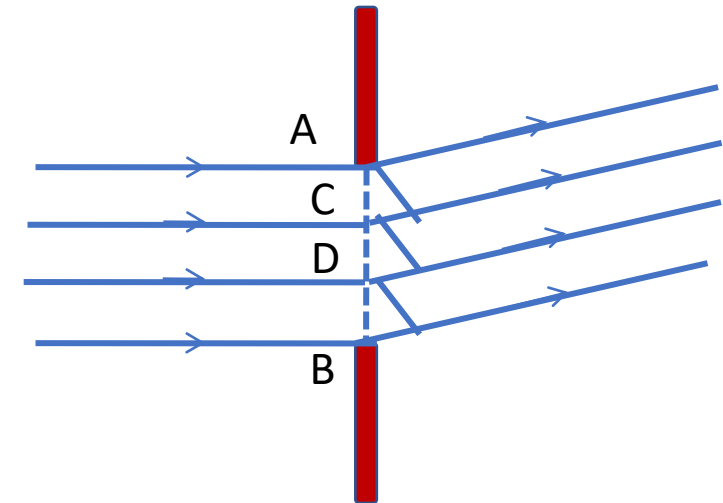
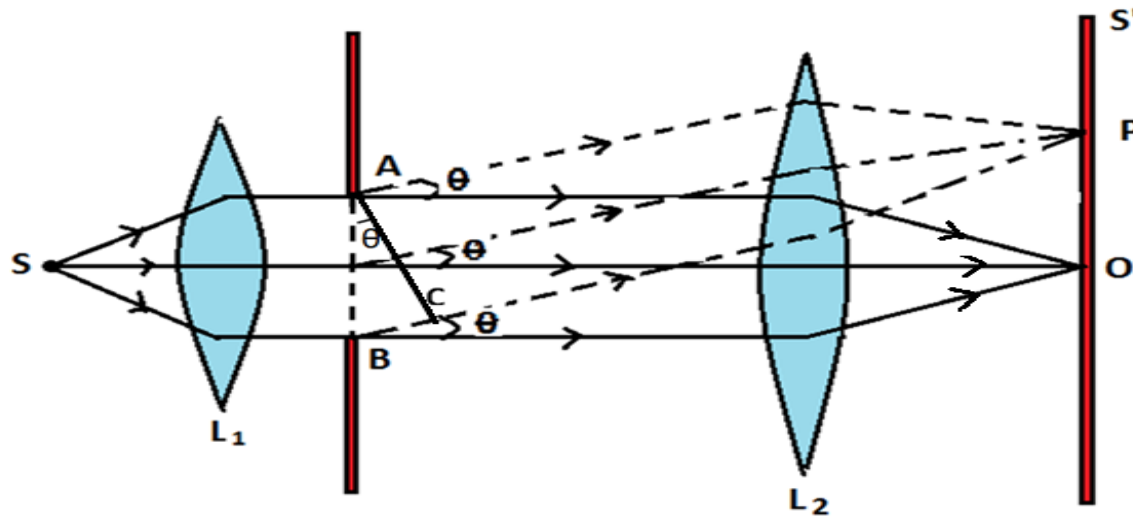
This equation gives the phase difference between the extreme secondary wavelets starting from A and B.



Fraunhofer's Diffraction at a Single Slit....

Let the width $AB = e$ be divided into p equal parts, then phase difference between wavelets from two consecutive slits will be $\frac{\delta}{p}$.

Let amplitude of each secondary wavelet is assumed to be ' a ', since each part is considered to have same width (e/p).

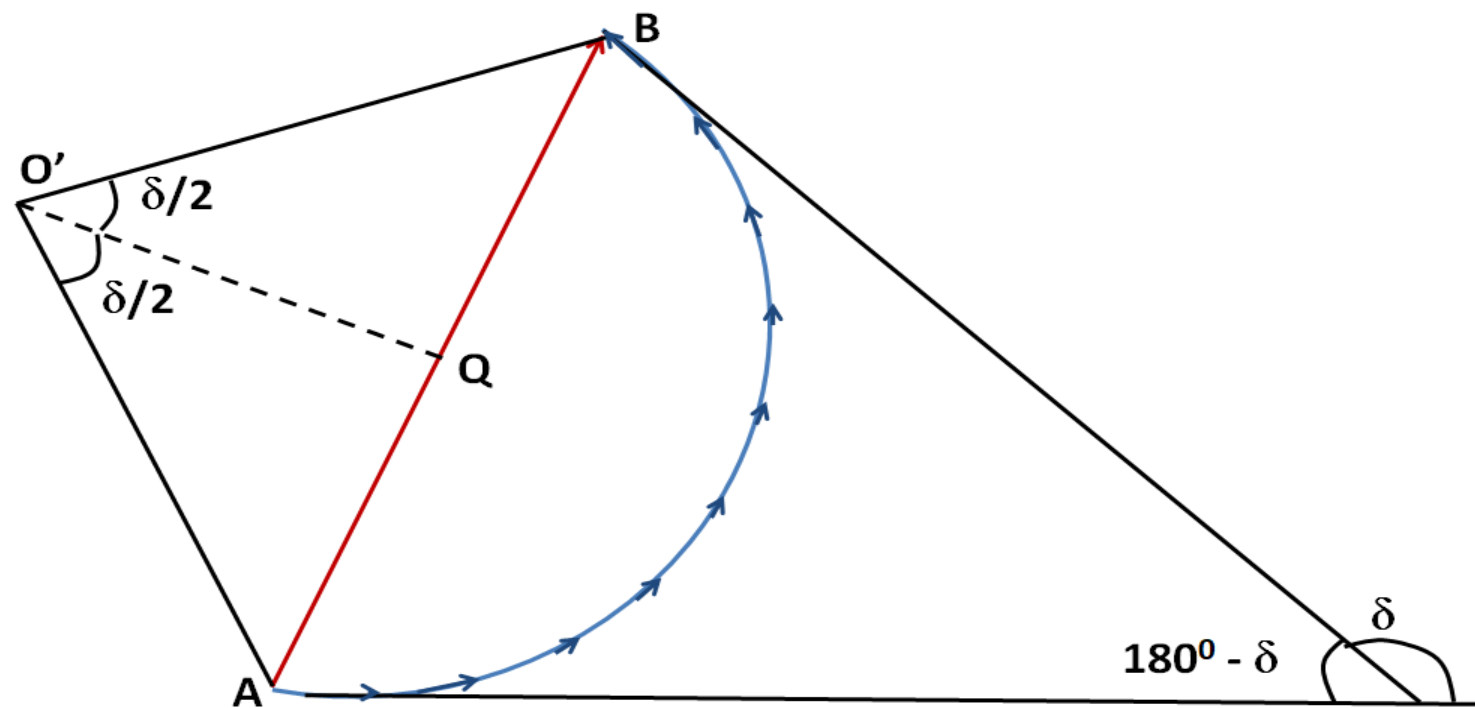
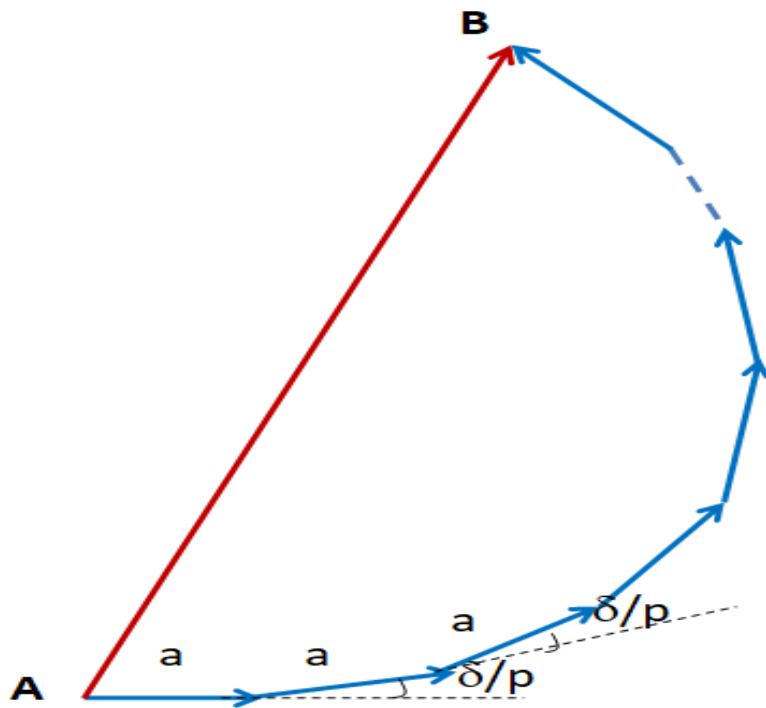


$AC=CD=DB=e/3$

Path difference between wavefronts from A and C $= e \sin \theta / 3$ and phase difference $= \frac{2\pi}{\lambda} \times \frac{e}{3} \sin \theta = \frac{\delta}{3}$

Fraunhofer's Diffraction at a Single Slit....

Vector polygon method or Phasor diagram method



Fraunhofer's Diffraction at a Single Slit....

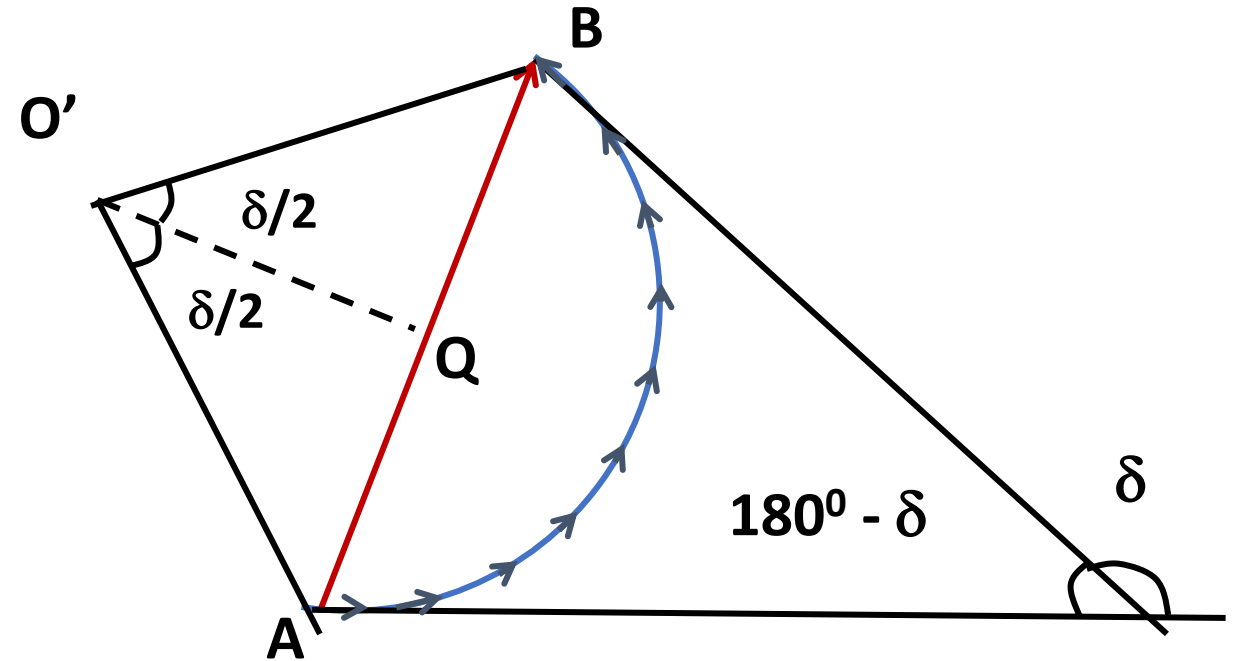
The line AB will represent the resultant disturbance. Since the number of secondary wavelets is very large then this step curve can be considered as a continuous arc. The phase difference between the wavelets from extreme ends of the slit AB is the angle between the tangent at point A and B.

$$\text{Angle } \delta = \frac{\text{arc}}{\text{radius}} = \frac{\text{arc}AB}{O'A}$$

As arc $AB = pa$ and $O'A = O'B = r$, so

$$\delta = \frac{pa}{r} \quad \dots(3)$$

$$\Rightarrow r = \frac{pa}{\delta} \quad \dots(4)$$



Fraunhofer's Diffraction at a Single Slit....

Let us draw a perpendicular $O'Q$ from the centre of the arc AB on chord which divides the line AB into two equal parts.

Here the resultant amplitude $AB=2AQ$... (5)

And

$$\angle AO'Q = \angle BO'Q = \frac{\delta}{2} \quad \dots (6)$$

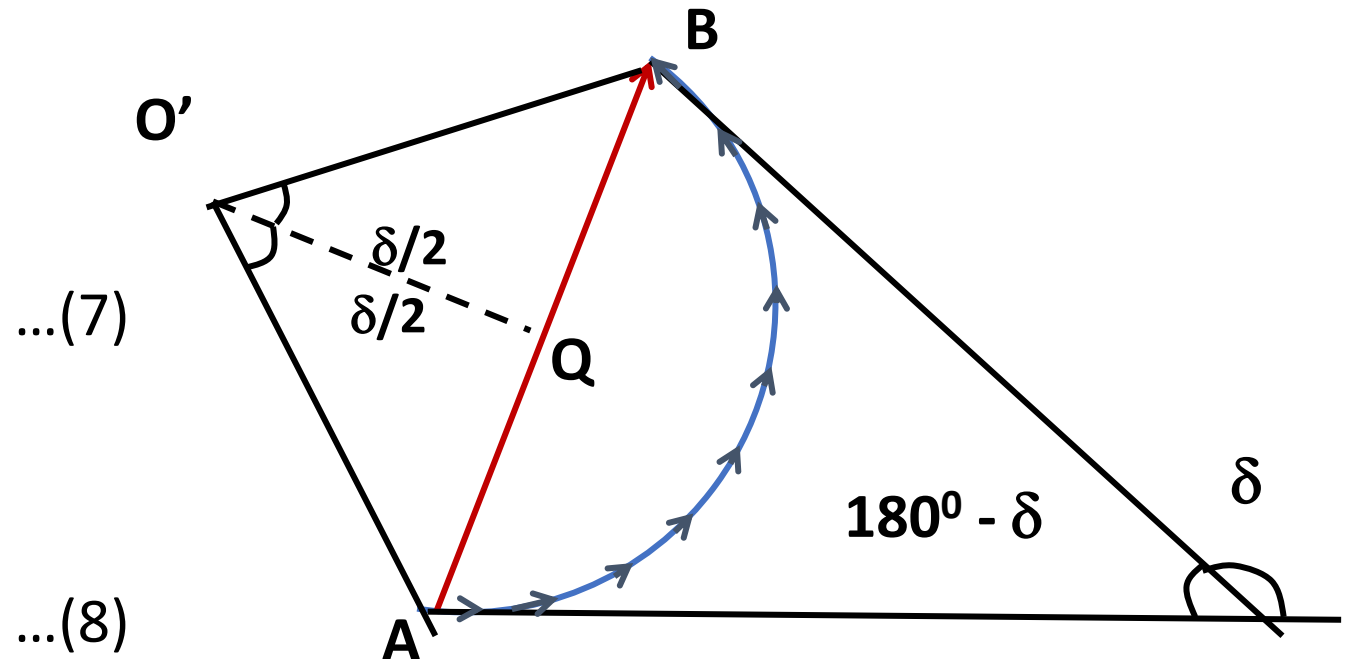
From $\triangle AO'Q$

$$\sin \frac{\delta}{2} = \frac{AQ}{O'A}$$

$$\Rightarrow AQ = O'A \sin \frac{\delta}{2} \quad \dots (7)$$

Using equation (7) in equation (5), we have

$$AB = 2AQ = 2O'A \sin \frac{\delta}{2} \quad \dots (8)$$



Fraunhofer's Diffraction at a Single Slit....

But $O'A=r$, so

$$AB = 2r \sin \frac{\delta}{2} \quad \dots(9)$$

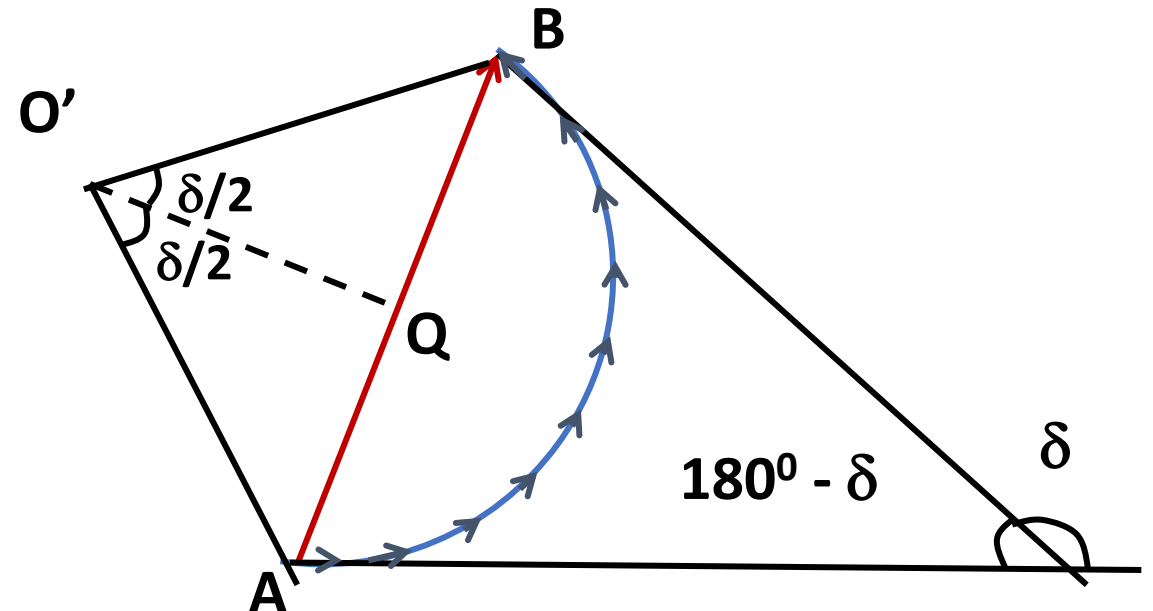
Substituting value of r from equation (4) in equation (9), we have

$$AB = 2 \frac{pa}{\delta} \sin \frac{\delta}{2} \quad \because r = \frac{pa}{\delta}$$

$$\text{Or } AB = pa \frac{\sin \frac{\delta}{2}}{\frac{\delta}{2}} \quad \dots(10)$$

Let R be the resultant amplitude, so

$$AB = R = pa \frac{\sin \frac{\delta}{2}}{\frac{\delta}{2}} \quad \dots(11)$$



Fraunhofer's Diffraction at a Single Slit....

From equation (2)

$$\alpha = \frac{\pi}{\lambda} e \sin \theta = \frac{\delta}{2} \quad \dots(12)$$

From equations (11) and (12), we have

$$\Rightarrow R = pa \frac{\sin \alpha}{\alpha}$$

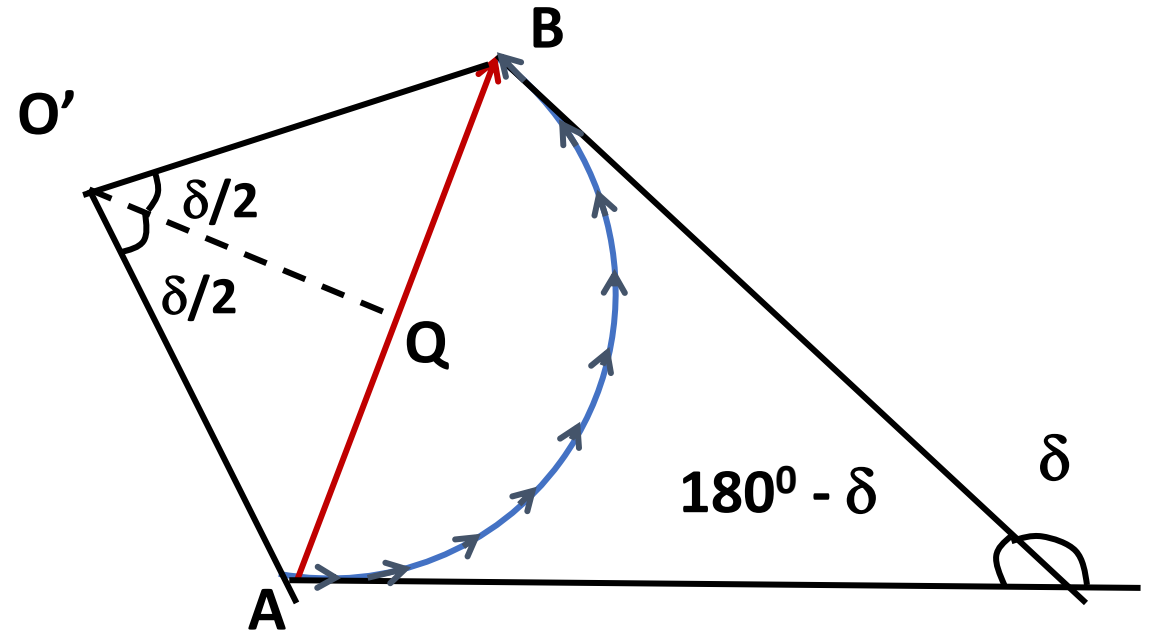
As we know that Intensity is proportional to (Amplitude)², so the resultant intensity

$$\Rightarrow I \propto \left(pa \frac{\sin \alpha}{\alpha} \right)^2$$

$$\Rightarrow I = kp^2 a^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

Let $kp^2 a^2 = I_0$

$$\Rightarrow I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad \dots(13)$$



This equation gives the resultant intensity at point P due to diffraction at a single slit.

Intensity Distribution in a Single Slit Diffraction Pattern

(i) Intensity and Position of Central (Principal) Maxima:

For central maxima (at point O), all the rays should strike directly (without diffraction) at point O on screen i.e. $\theta=0$.

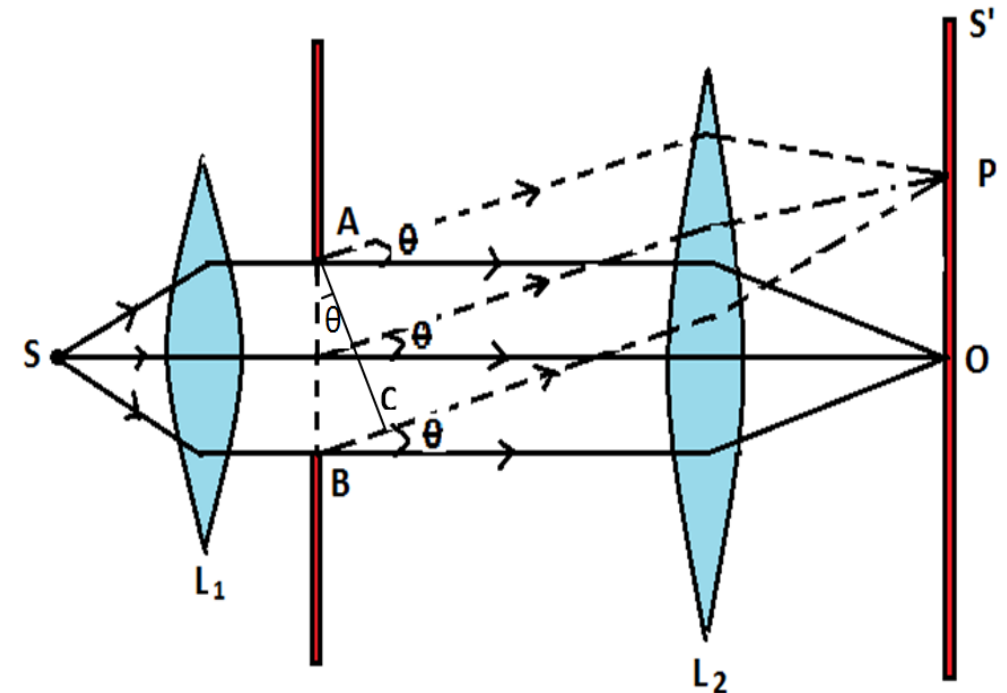
$$\text{So, } \alpha = \frac{\pi}{\lambda} e \sin \theta = 0 \quad \dots(14)$$

$$\text{As } \lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} = 1 \quad \dots(15)$$

From equations (13) and (15), we have

$$I_{\max} = I_o \quad \dots(16)$$

i.e. if the angle of diffraction is zero then maximum intensity will be observed on the screen.



Intensity Distribution in a Single Slit Diffraction Pattern...

(ii) Intensity and Position of Minima

- From equation (13) it is clear that for minimum intensity

$$I_{\min} = 0 \Rightarrow \frac{\sin \alpha}{\alpha} = 0$$

But $\alpha \neq 0$

- For $I = 0$ (minimum) ... (17)

$$\sin \alpha = 0$$

- $\Rightarrow \alpha = \pm n\pi$... (18)

$$\Rightarrow \alpha = \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi, \dots$$

Here $n=1,2,3,\dots$ except zero because for $n = 0, \alpha \rightarrow 0$ which corresponds to the condition for maximum intensity or principal maxima.

Intensity Distribution in a Single Slit Diffraction Pattern...

- So from equations (12) and (14), we have

$$\alpha = \frac{\pi}{\lambda} e \sin \theta = \pm n \pi$$

- $\Rightarrow e \sin \theta = \pm n \lambda \quad \dots(19)$
- Here $\pm$ sign indicates minima on either side of principal maxima.

(iii) Intensity and Position of Secondary Maxima

So, in the diffraction pattern we have a central maximum and on either side of central maximum we have different orders of minima. In between these minima there are secondary maxima of weak intensity.

Intensity Distribution in a Single Slit Diffraction Pattern...

In order to find the position of secondary maxima differentiate equation (13) with respect to α and equate it to zero.

$$\text{i.e. } \frac{dI}{d\alpha} = 0 \quad \dots(20)$$

$$\Rightarrow \frac{d}{d\alpha} \left[I_o \left(\frac{\sin \alpha}{\alpha} \right)^2 \right] = 0$$

$$\Rightarrow 2I_o \left(\frac{\sin \alpha}{\alpha} \right) \left[\frac{\alpha \cos \alpha - \sin \alpha \cdot 1}{\alpha^2} \right] = 0 \quad \dots(21)$$

Here $I_o \neq 0$

$\frac{\sin \alpha}{\alpha} \neq 0$ (because $\frac{\sin \alpha}{\alpha} = 0$ corresponds to minima)

$\alpha \neq 0$ (because $\alpha = 0$ corresponds to Principal maximum)

Intensity Distribution in a Single Slit Diffraction Pattern...

Hence from equation (21), we have

$$\alpha \cos \alpha - \sin \alpha = 0$$

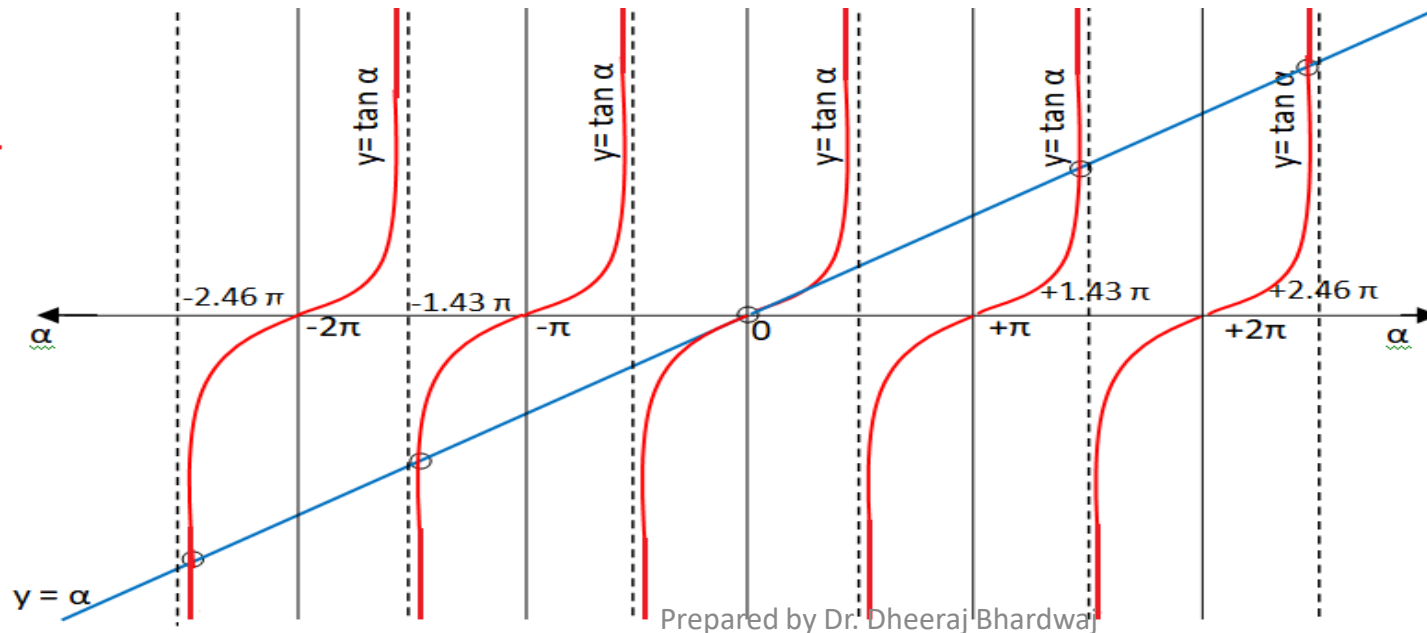
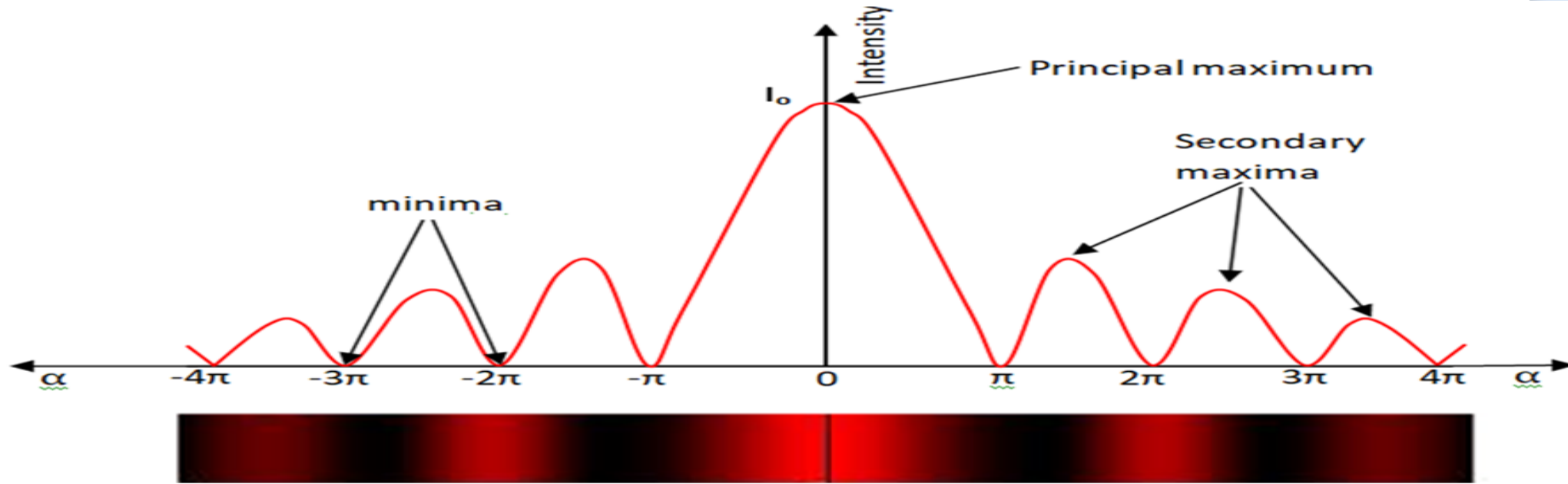
$$\Rightarrow \alpha = \tan \alpha \quad \dots(22)$$

The solution of equation (22) will give position of secondary maxima. This equation can be solved by plotting $y = \alpha$ and $y = \tan \alpha$ curves on a single graph.

The equation $y = \alpha$ represents a straight line passing through the origin and makes an angle of 45° from the x-axis and the equation $y = \tan \alpha$ are the discontinuous curves from $y = -\infty$ to $y = +\infty$.

The points of interaction of these two curves will give the value of α which corresponds to position of secondary maxima.

Intensity Distribution in a Single Slit Diffraction Pattern...



Intensity Distribution in a Single Slit Diffraction Pattern...

- The approximate values of α are $\alpha = 0, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots, \pm \frac{(2n+1)\pi}{2}$

and exact values of α are $\alpha = 0, \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \dots$

For the secondary maxima the values of α are

$$\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots, \pm \frac{(2n+1)\pi}{2}$$

$$\alpha = \frac{\pi}{\lambda} e \sin \theta = \pm \frac{(2n+1)\pi}{2}$$

$$\Rightarrow e \sin \theta = \pm \frac{(2n+1)\lambda}{2}$$

Where $\alpha=0$ corresponds to central maxima.

Intensity Distribution in a Single Slit Diffraction Pattern...

Let us substitute different values of α in equation (13), to get the intensities of different orders of secondary maxima.

For central maximum $\alpha = 0$

$$I_{\max} = I_o$$

...(23)

For first secondary maxima $\alpha = \pm \frac{3\pi}{2}$

$$I = I_o \left(\frac{\sin \frac{3\pi}{2}}{\frac{3\pi}{2}} \right)^2 = I_o \left(\frac{-1}{\frac{3\pi}{2}} \right)^2$$

$$\Rightarrow I_1 = \frac{4I_o}{9\pi^2} \approx \frac{I_o}{22} \quad \text{...(24)}$$

Intensity Distribution in a Single Slit Diffraction Pattern...

For second secondary maxima

$$\alpha = \pm \frac{5\pi}{2}$$
$$I = I_0 \left(\frac{\sin \frac{5\pi}{2}}{\frac{5\pi}{2}} \right)^2 = I_0 \left(\frac{1}{\frac{5\pi}{2}} \right)^2$$
$$\Rightarrow I_1 = \frac{4I_0}{25\pi^2} \approx \frac{I_0}{61}$$

...(25)

For third secondary maxima

$$\alpha = \pm \frac{7\pi}{2}$$
$$I = I_0 \left(\frac{\sin \frac{7\pi}{2}}{\frac{7\pi}{2}} \right)^2 = I_0 \left(\frac{-1}{\frac{7\pi}{2}} \right)^2$$
$$\Rightarrow I_1 = \frac{4I_0}{49\pi^2} \approx \frac{I_0}{121}$$

...(26)

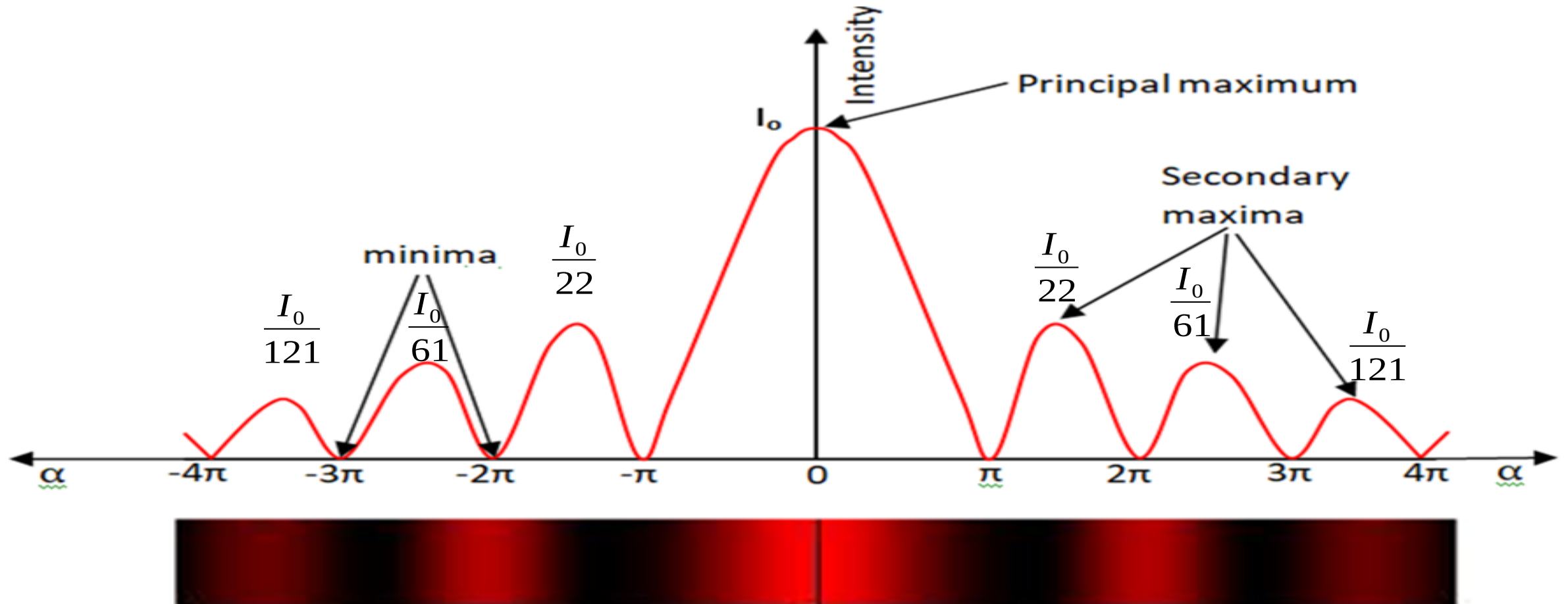
Intensity Distribution in a Single Slit Diffraction Pattern...

So the ratio of intensities of maxima in Fraunhofer diffraction at single slit is given as

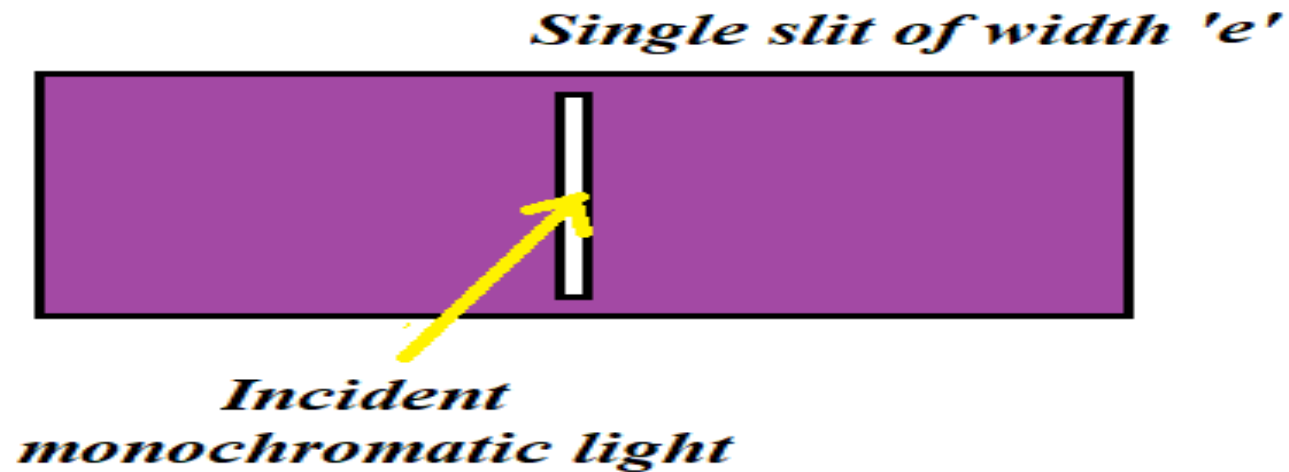
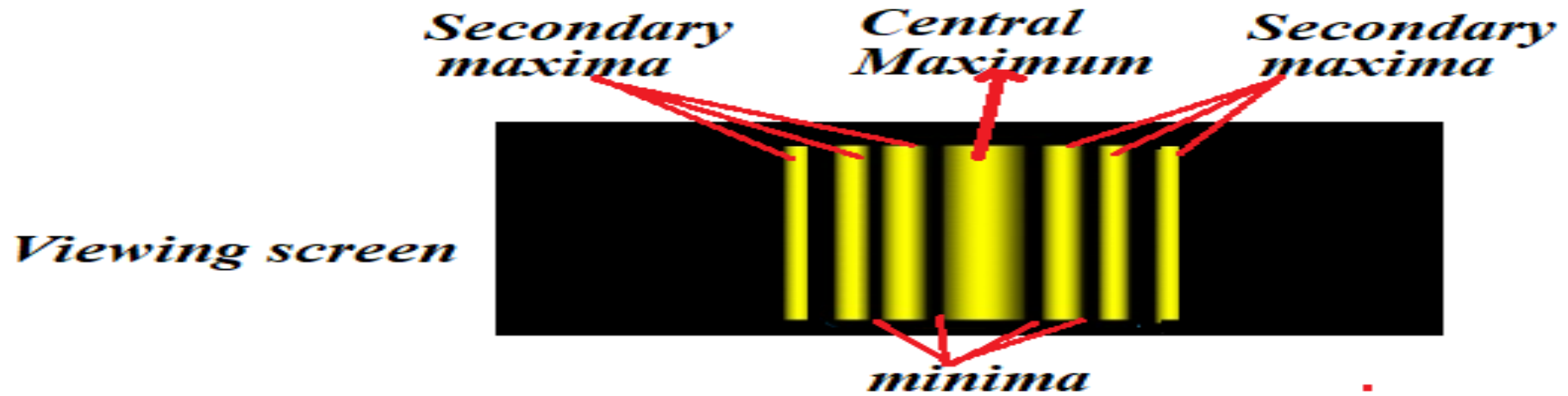
$$I_0 : \frac{4I_0}{9\pi^2} : \frac{4I_0}{25\pi^2} : \frac{4I_0}{49\pi^2} : \dots$$
$$\Rightarrow I_0 : \frac{I_0}{22} : \frac{I_0}{61} : \frac{I_0}{121} : \dots$$

So it is clear that the first secondary maxima on either side of the central maximum have only about 4.5% of the height of the central maximum and the intensity decreases as the order of secondary maxima increases. It is also clear that secondary maxima does not fall exactly half way between minima, they are slightly displaced towards central maximum.

Intensity Distribution in a Single Slit Diffraction Pattern...



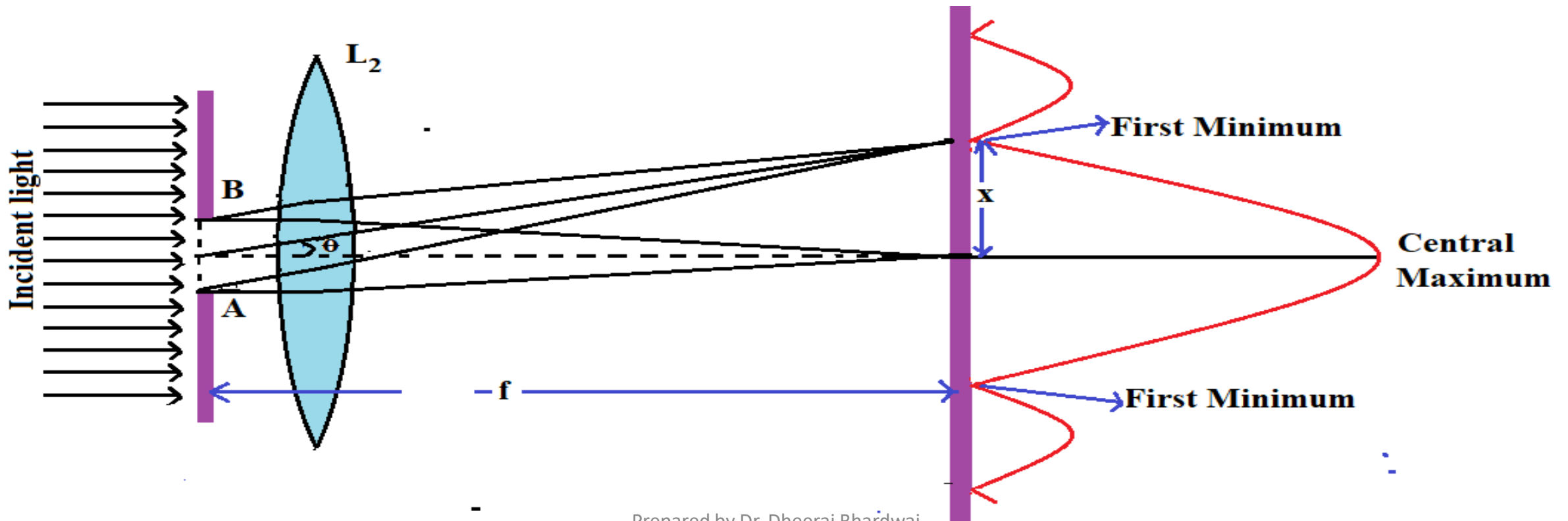
Single Slit Diffraction Pattern



Single Slit Diffraction

Width of central maximum:

Width of central maximum is defined as the distance between the first minima on either side of central maximum.



Width of central maximum

If the distance between the central maximum and the first minimum is 'x' then

$$\sin \theta \approx \tan \theta \approx \frac{x}{f}$$

where 'f' is the focal length of the convex lens. Here it is assumed that convex lens is very close to the slit and screen is at its focal plane. For first minimum

$$\sin \theta = \pm \frac{\lambda}{e}$$

Also

$$\sin \theta = \frac{x}{f} \\ \Rightarrow \frac{x}{f} = \pm \frac{\lambda}{e} \Rightarrow x = \pm \frac{\lambda f}{e}$$

∴ The width of central maximum

$$W = 2x = \frac{2\lambda f}{e}$$

The angular width of central maximum

$$2\theta = \frac{2x}{f} = \frac{2\lambda}{e}$$

Width of central maximum

$$W = 2x = \frac{2\lambda f}{e}$$

Thus it is clear that width of the central maximum is proportional to the wavelength of incident light ($W \propto \lambda$) and inversely proportional to the width of the slit ($W \propto \frac{1}{e}$).

- $W \propto \lambda$ and as $\lambda_{\text{red}} > \lambda_{\text{violet}}$. So, width of central maximum for red light will be more than that for violet i.e. $W_{\text{red}} > W_{\text{violet}}$ or we can say that red light is diffracted more.
- If the white light is used, the central maximum is white in the middle but reddish on its outer edge, shading into purple and other impure colours farther out.

Width of central maximum

As $W \propto \frac{1}{e}$

- The narrower the slit, the broader the central maximum; the central maximum and minima are distinct on both sides.
- If the slit is wide, the central maximum is narrow and the pattern shrinks rapidly to a smaller scale.
- If the width of slit is reduced to the extent that it is equal to λ ($e=\lambda$) then

$\sin \theta = \pm 1$ or $\theta = 90^\circ$, i.e., the central maximum spreads over the whole screen.

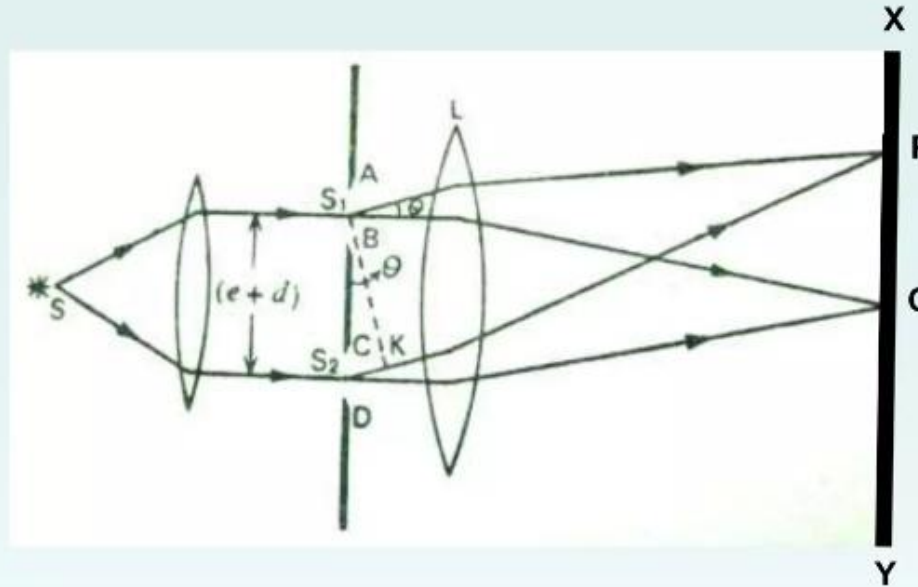
➤ Width of central maximum is twice that of any other maxima.

Fraunhofer's Diffraction at a Double Slit

Slit width = e

Slit separation / width of Opaque space = d

Distance between two successive slits = $e + d$



- * Let a parallel beam of monochromatic light of wavelength λ be incident normally upon two parallel slits AB and CD , each of width e , separated by opaque space d .
- * The distance between the corresponding points of the two slits is $(e+d)$.
- * Let the diffracted light be focused by a convex lens L on a screen XY placed in the focal plane of the lens.

Continued....



The pattern obtained on the screen is the diffraction pattern due to a single slit on which a system of interference fringes is superposed.

By Huygen's principle, every point in the slits AB and CD sends out secondary wavelets in all directions.

From the theory of diffraction at a single slit, the resultant amplitude due to wavelets diffracted from each slit in a direction θ is

$$\frac{A \sin \alpha}{\alpha}$$

Where A is a constant and $\alpha = \frac{\pi e \sin \theta}{\lambda}$

We can, therefore, consider the two slits as equivalent to two coherent sources placed at the middle points S_1 and S_2 of the slits, and each sending a wavelet of amplitude $(A \sin \alpha)/\alpha$ in the direction of θ .

Continued....



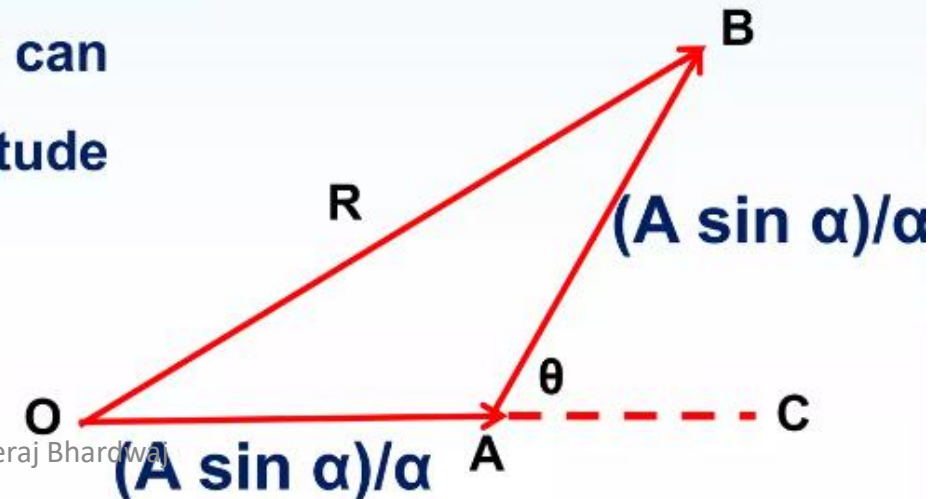
Therefore, the resultant amplitude at a point P on the screen will be the result of interference between two waves of same amplitude $(A \sin \alpha)/\alpha$, and having a phase difference δ .

Let us draw S_1K perpendicular to S_2K . The path difference between the wavelets from S_1 and S_2 in the direction $\theta = S_2K = (e+d) \sin \theta$

$$\begin{aligned}\text{Hence the phase difference is } \delta &= (2\pi/\lambda) \times \text{path difference} \\ &= (2\pi/\lambda) \times (e+d) \sin \theta\end{aligned}$$

The resultant amplitude R at P can be determined by the vector amplitude diagram, which gives

$$OB^2 = OA^2 + AB^2 + 2(OA)(AB) \cos BAC$$



Continued....

$$OB^2 = OA^2 + AB^2 + 2(OA)(AB) \cos BAC$$

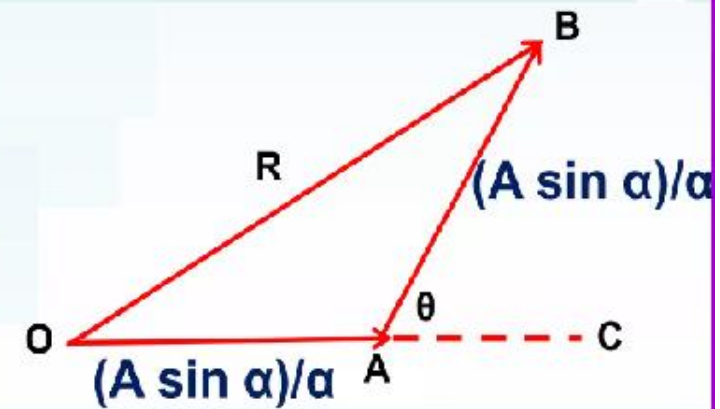
$$R^2 = A \left(\frac{\sin \alpha}{\alpha} \right)^2 + A \left(\frac{\sin \alpha}{\alpha} \right)^2 + 2 \left(\frac{\sin \alpha}{\alpha} \right) \left(\frac{\sin \alpha}{\alpha} \right) \cos \delta$$

$$R^2 = \left(\frac{A \sin \alpha}{\alpha} \right)^2 (2 + 2 \cos \delta)$$

$$= \frac{A^2 \sin^2 \alpha}{\alpha^2} 4 \cos^2 \frac{\delta}{2}$$

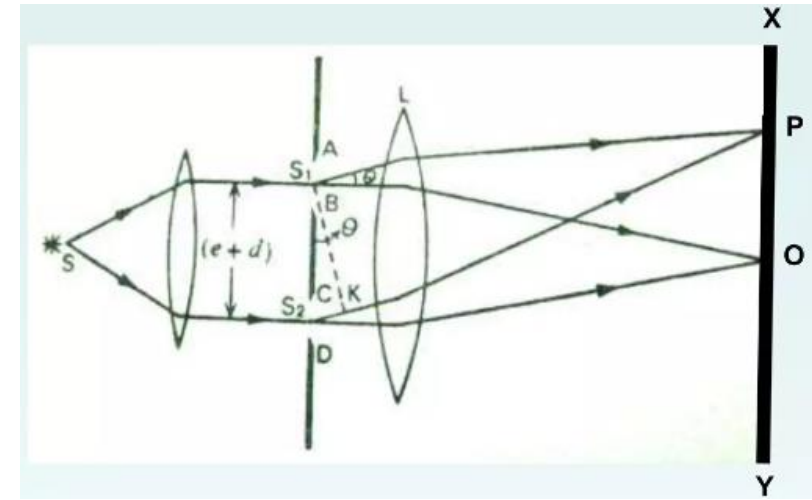
$$= 4A^2 \frac{\sin^2 \alpha}{\alpha^2} \cos^2 \beta$$

$$\text{where } \beta = \frac{\delta}{2} = \frac{\pi}{\lambda} (e + d) \sin \theta$$



Therefore, the resultant intensity at P is $I = R^2 = 4A^2 \frac{\sin^2 \alpha}{\alpha^2} \cos^2 \beta$

Physical Significance of Double-Slit Diffraction:-



The resultant intensity at P is $I = R^2 = 4A^2 \frac{\sin^2 \alpha}{\alpha^2} \cos^2 \beta$

$$\text{where } \beta = \frac{\delta}{2} = \frac{\pi}{\lambda} (e + d) \sin \theta$$

$$\alpha = \frac{\pi e \sin \theta}{\lambda}$$

Slit width = e

Slit separation / width of Opaque space = d

Distance between two successive slits = $e + d$

Thus the intensity in the resultant pattern depends on two factors:

(i) $(\sin^2 \alpha) / \alpha^2$, which gives the diffraction pattern due to each individual slit

(ii) $\cos^2 \beta$, which gives interference pattern due to diffracted light waves from the two slits.

Continued....



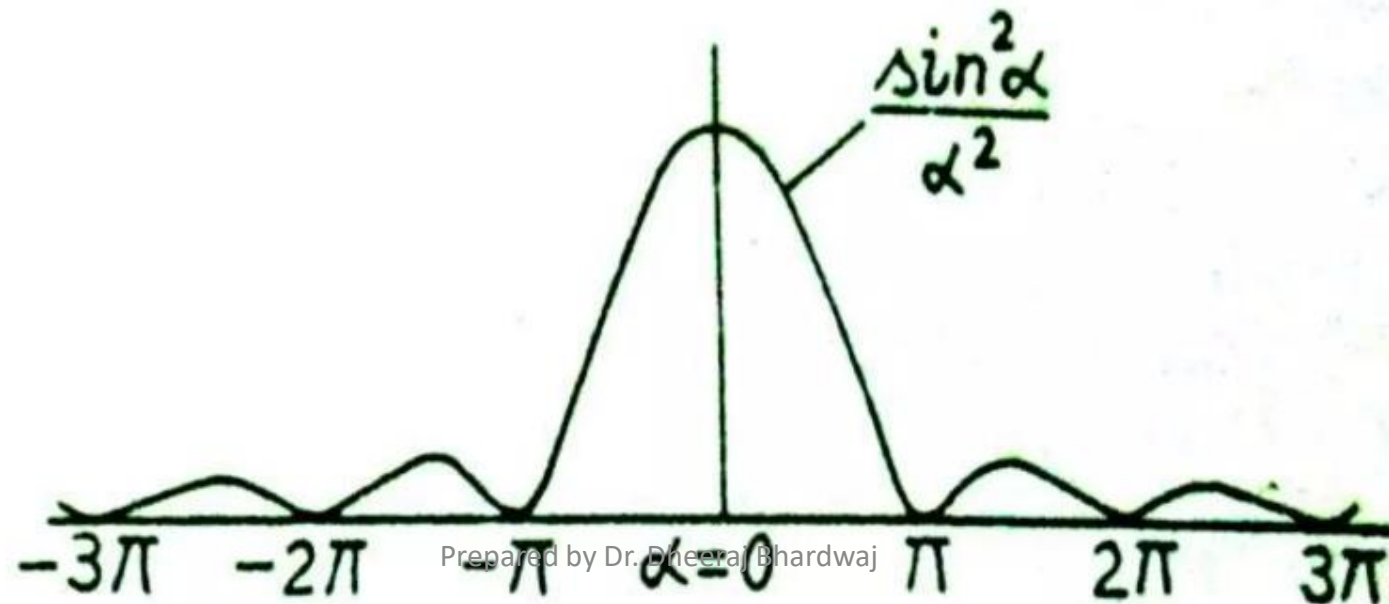
(a) The diffraction term, $(\sin^2 \alpha) / \alpha^2$, gives a central maximum in the direction $\theta = 0$, having alternately minima and subsidiary maxima of decreasing intensity on either side.

The minima are obtained in the directions given by

$$\sin \alpha = 0, \quad \alpha = \pm m\pi$$

$$(\pi e \sin \theta) / \lambda = \pm m\pi$$

$$e \sin \theta = \pm m\pi \quad (m=1,2,3,\dots, \text{except zero})$$



Continued....

(b) The interference term, $\cos^2 \beta$, gives a set of equidistant dark and bright fringes, as in Young's double-slit interference experiment.

The bright fringes (maxima) are obtained in the directions given by

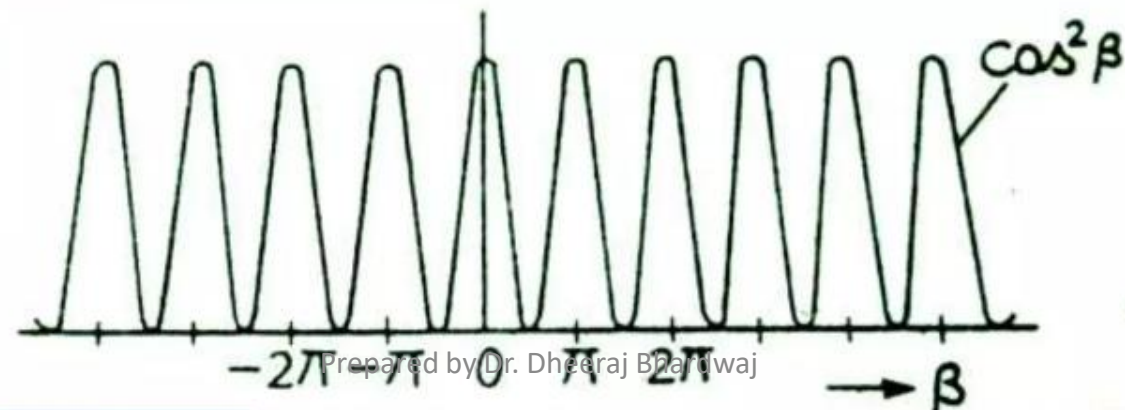
$$\cos^2 \beta = 1$$

$$\beta = \pm n\pi$$

$$(\pi/\lambda) (e+d) \sin \theta = \pm n\pi$$

$$(e+d) \sin \theta = \pm n\lambda, \quad \text{Where } n = 0, 1, 2, \dots$$

The various maxima corresponding to $n = 0, 1, 2, \dots$ are zero-order, first-order, second-order maxima

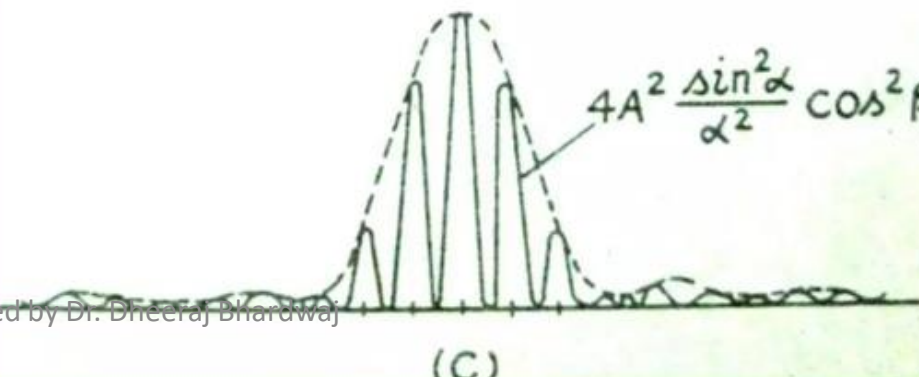
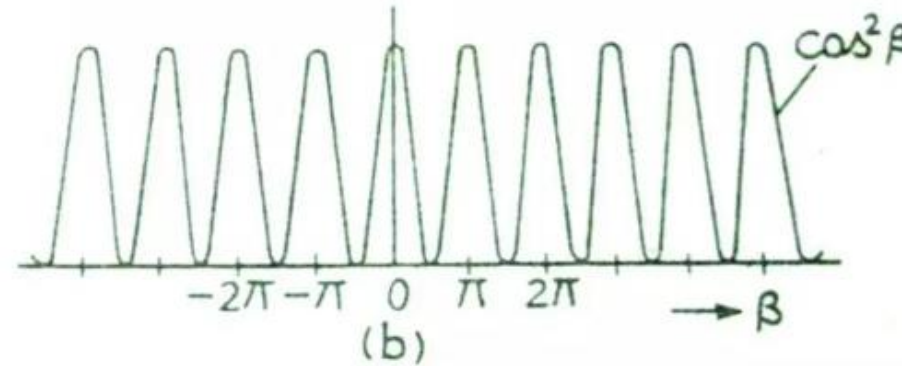
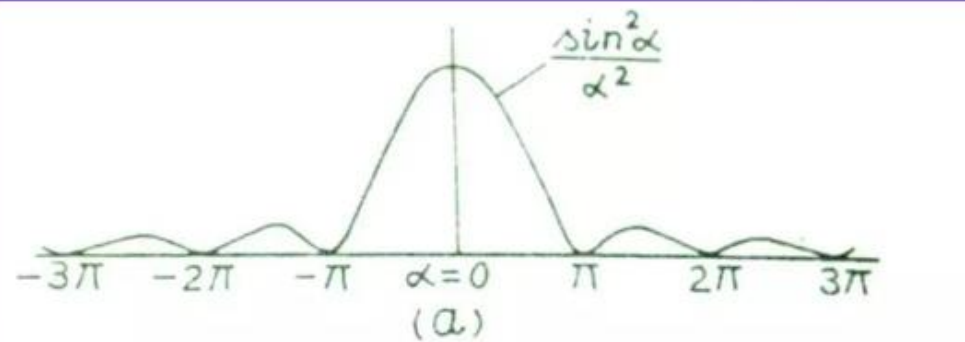


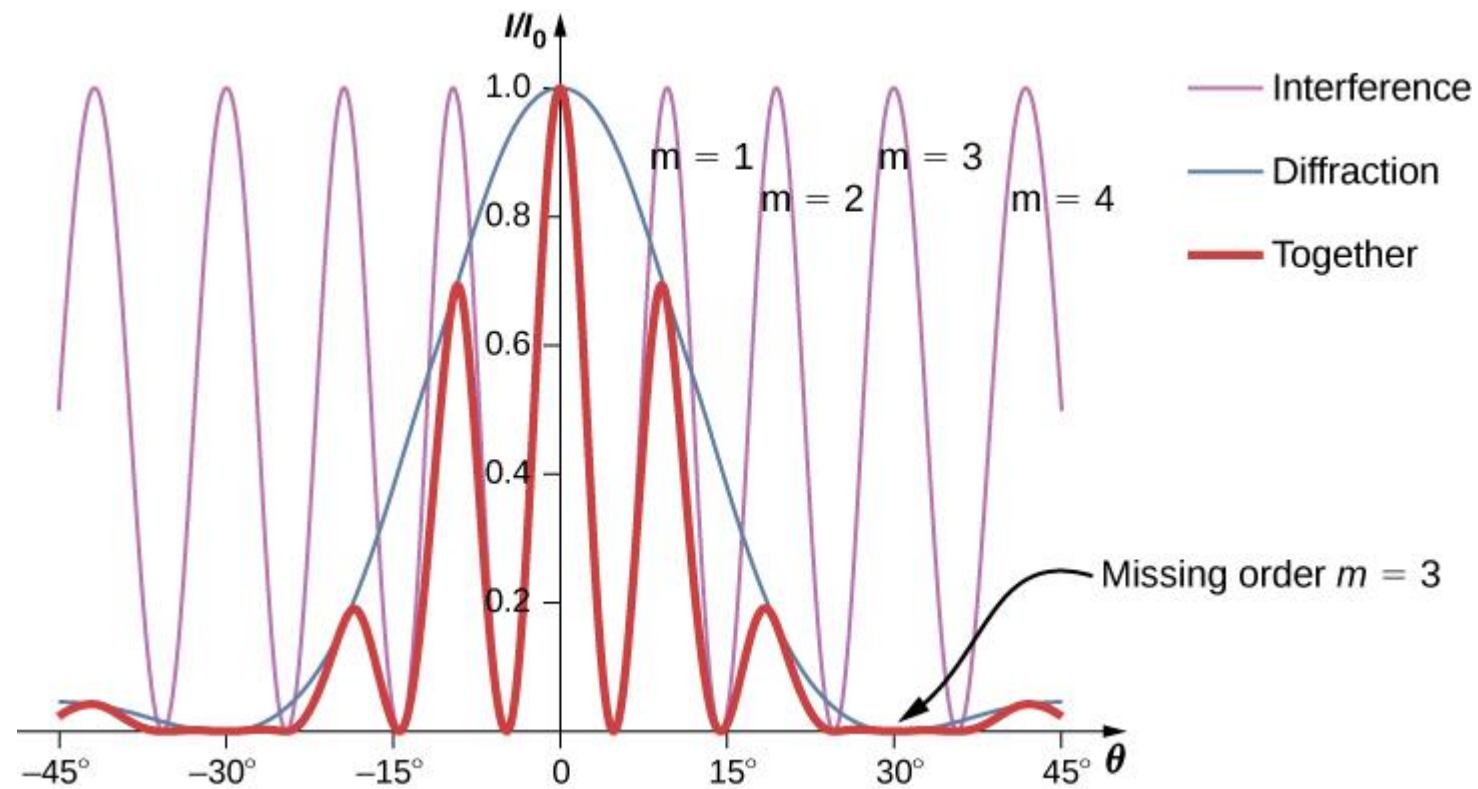
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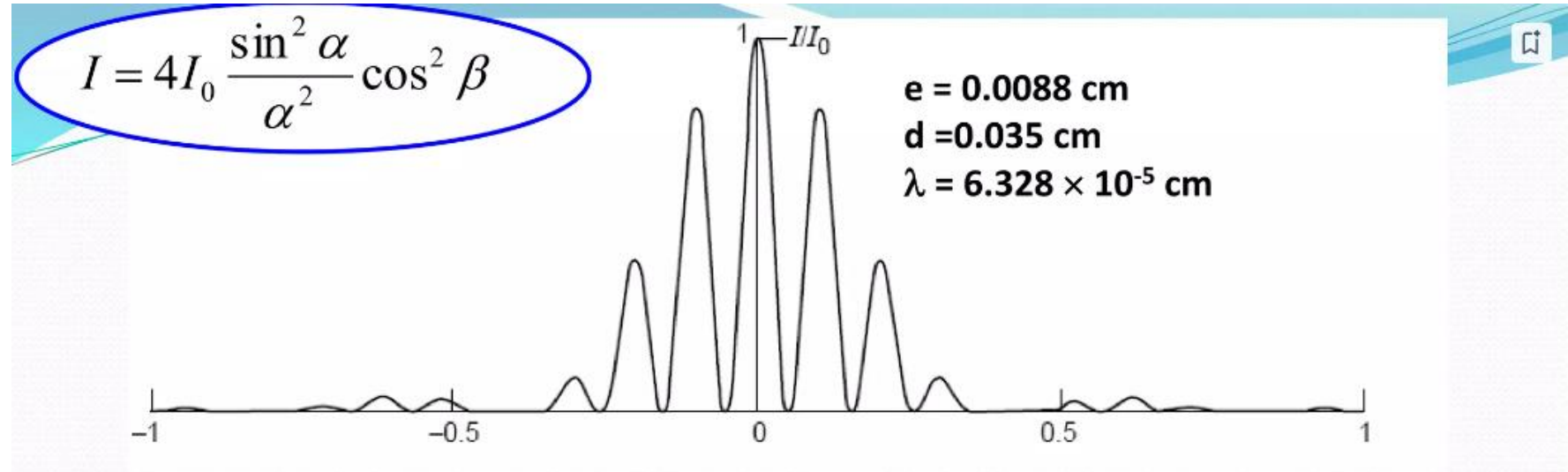
(c) The intensity distribution in the resultant diffraction pattern is a plot of the product of the constant term $4 A^2$, diffraction term $(\sin^2 \alpha)/\alpha^2$, and the interference term $\cos^2 \beta$.

The entire pattern may be considered as consisting interference fringes due to light from both slits, the intensities of these fringes being governed by the diffraction occurring at the individual slits.





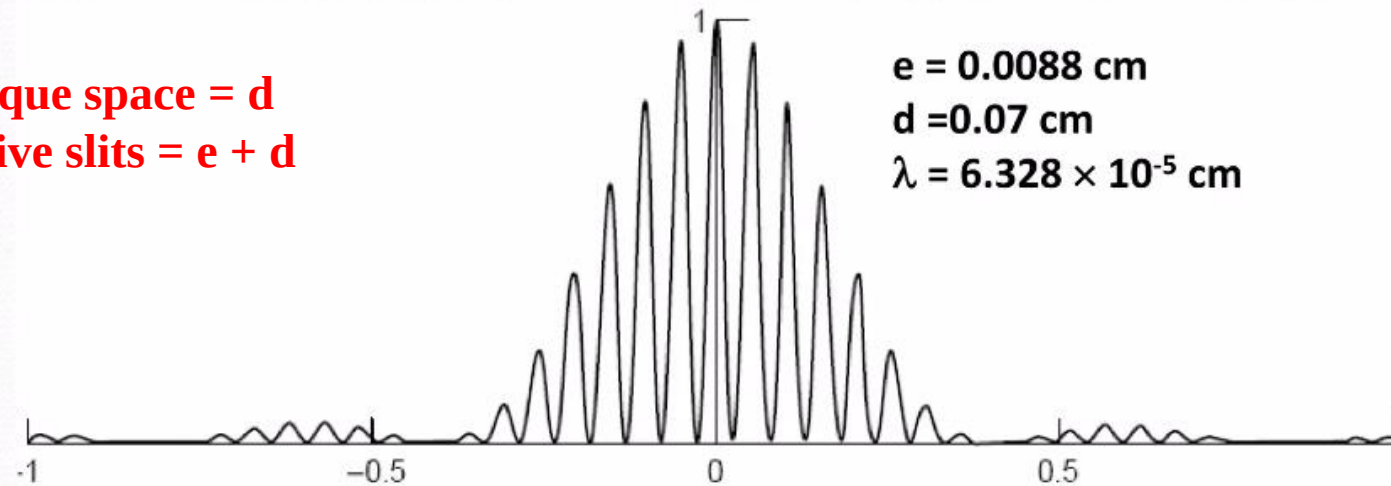
Effect of width of Opaque space (d) on of double slit diffraction Pattern :-



Slit width = e

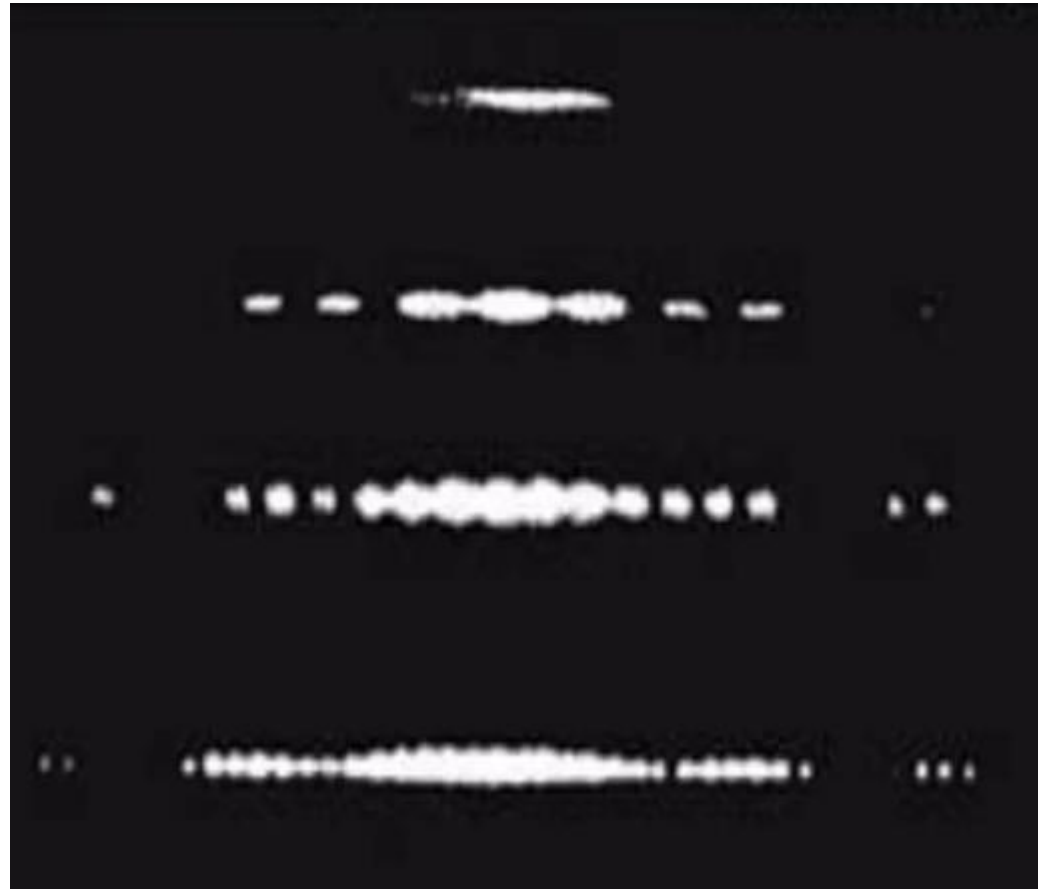
Slit separation / width of Opaque space = d

Distance between two successive slits = e + d



The double-slit intensity distribution corresponding to $e = 0.0088$ cm, $\lambda = 6.328 \times 10^{-5}$ cm, and $d = 0.035$ cm and $d = 0.07$ cm respectively.

The double-slit Fraunhofer diffraction pattern corresponding to $e = 0.0088$ cm and $\lambda = 6.328 \times 10^{-5}$ cm. The values of d are 0, 0.0176, 0.035, and 0.070 cm, respectively.



Questions

- Q1. What do you mean by Diffraction of Light? What is condition for diffraction?
- Q2. Distinguish between Fresnel's and Fraunhofer's classes of diffraction.
- Q3. What will happen to the diffraction pattern, if slit is made narrower or wider?
- Q4. Sketch the diffraction pattern formed by the Single slit. Write the factors on which angular width of central maximum of a single slit diffraction pattern depend upon.
- Q5. What happens to the width of the central maximum in a single-slit diffraction if the slit width is increased?

Questions

- Q6. What do you mean by diffraction? Distinguish between Fresnel's and Fraunhofer's classes of diffraction.
- Q7. Find out an expression for intensity distribution due to Fraunhofer diffraction at a single slit. Draw the intensity distribution curve. Show that the intensity of the first subsidiary maxima is about 4.5% of that of the principal maxima.
- Q8. Discuss the phenomenon of Fraunhofer diffraction at single slit and show that the relative intensities of successive maxima are nearly:

$$I_o : \frac{4I_o}{9\pi^2} : \frac{4I_o}{25\pi^2} : \frac{4I_o}{49\pi^2} : \dots$$

- Q9. Find out the width of central maximum in the intensity distribution for Fraunhofer diffraction by single slit.

**Thank
You**

